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Global Technology

MLCC: Continued Demand Growth for High-Value-added Products for AI & Data Centers

Remain OW on Murata, SEMCO & Yageo, which we see as MLCC demand growth beneficiaries, and upgrade Taiyo Yuden to EW.

Key Takeaways

- We forecast global MLCC shipment value to expand at a CAGR of 20.3%, driven by growing demand for AI servers/DCs.
- MLCCs are a key business, accounting for 50% of sales at Murata Mfg, 46% at SEMCO, 19% at Yageo, and 70% at Taiyo Yuden in 2025.
- We expect Murata, the top player by global market share, and No. 2 SEMCO to benefit particularly strongly from expanding demand for AI server/DC MLCCs.



Tech Diffusion

A Morgan Stanley Research
Key Theme of 2026

MLCC is early cycle in AI server & rack penetration:

Revenue growth is inflecting, driven by a structural change from AI-related MLCC content. This shift to higher-margin AI products will support sustained earnings growth worth more than cyclical earnings growth – and thus, higher valuations. In short, MLCC is perhaps the new investment highly levered to data center spending among the "AI picks & shovels".

Demand in high-value-added MLCCs for AI/DC to

keep growing from 2027: We forecast CAGR in global MLCC shipment value at 20.3% in 2025-31 (\$14.67bn to \$44.45bn) based on our outlook for continued growth in sales of small, high-capacity, high-value-added MLCCs for AI servers/DCs.

Murata & SEMCO the biggest beneficiaries: MLCCs are a key business, with 50% of sales at Murata, 46% SEMCO, 19% Yageo, and 70% Taiyo Yuden in 2025. We expect growing demand for AI server/DC MLCCs to be particularly positive for Murata (top player at 41% global MLCC market share in 2025) and SEMCO (#2 at 23%). We view the sell-off in these stocks from their peaks as a re-entry opportunity.

Remain OW on Murata as Top Pick in Japan electronic components; raise Taiyo Yuden to EW: We raise Murata forecasts as in [Exhibit 25](#). However, we lower our DCF-based PT from ¥12,500 to ¥11,000, mainly reflecting an increase in our DCF model risk-free rate (2.6% to 3.0%). We also revise Taiyo Yuden estimates ([Exhibit 33](#)), but expect it to take longer than Murata & SEMCO to benefit from AI server/DC MLCC demand growth. We lower our DCF-based PT from ¥12,500 to ¥10,500

MORGAN STANLEY MUFG SECURITIES CO., LTD.+

Shoji Sato

Equity Analyst

Shoji.Sato@morganstanleymufg.com

+81 3 6836-8404

MORGAN STANLEY TAIWAN LIMITED+

Howard Kao

Equity Analyst

Howard.Kao@morganstanley.com

+886 2 2730-2989

MORGAN STANLEY & CO. INTERNATIONAL PLC+

Shawn Kim

Equity Analyst

Shawn.Kim@morganstanley.com

+44 20 7677-1018

MORGAN STANLEY MUFG SECURITIES CO., LTD.+

Sota Harashima

Equity Analyst

Sota.Harashima@morganstanleymufg.com

+81 3 6836-8897

MORGAN STANLEY ASIA LIMITED+

Duan Liu

Equity Analyst

Duan.Liu@morganstanley.com

+852 2239-7357

MORGAN STANLEY & CO. INTERNATIONAL PLC+

Cindy Huang

Equity Analyst

Cindy.Huang@morganstanley.com

+44 20 7425-2915

WHAT'S CHANGED

Taiyo Yuden (6976.T)	From	To
Rating	Underweight	Equal-weight
Price Target	¥12,500	¥10,500

Murata Manufacturing (6981.T)	From	To
Price Target	¥12,500	¥11,000

Yageo Corp. (2327.TW)	From	To
Price Target	NT\$1,465.00	NT\$1,050.00

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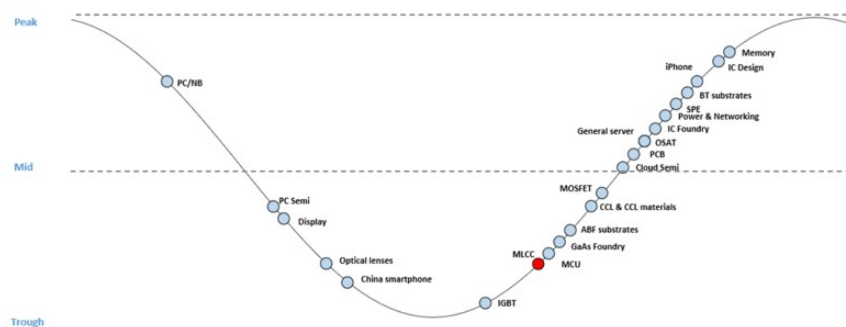
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Investment Perspective

MLCC is early cycle in AI server & rack penetration

There is immense change underway for the MLCC industry with an investment narrative that is changing as AI chips/server demand is increasingly driving MLCC scarcity. AI is shifting from semiconductors to MLCCs, creating a supply/demand mismatch in high-end components. The high-end AI MLCC market is effectively a Japanese-Korean duopoly with Murata and SEMCO together hold around 85% share. High-capacitance, low-ESL parts for AI accelerators require hundreds to over a thousand dielectric layers with near-zero defects and face strict customer qualification — the barrier preventing a repeat of 2017-18, when shortages invited entry and the cycle self-corrected. AI platforms now compete directly with consumer, automotive, and industrial applications for advanced materials, heightening supply risk for MLCC traditional customers. We are still in the early innings, and supply concerns are extending beyond near-term shortages to longer-term supply security. As a result, we are seeing an increasing number of long-term supply agreements being signed between suppliers and customers.

Exhibit 1: Sequencing the technology cycle – still early cycle for MLCC



Source: Morgan Stanley Research estimates

Turning point in MLCC pricing

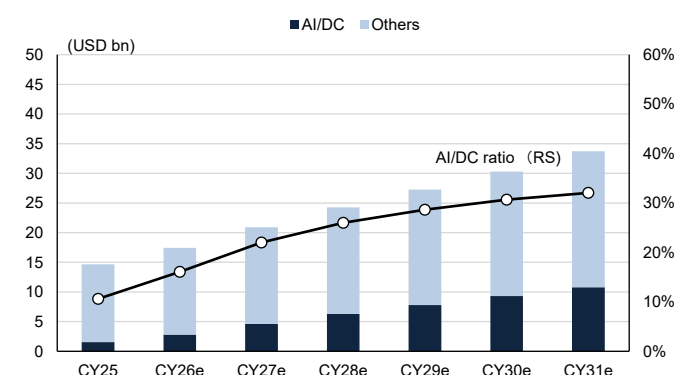
While Murata Mfg.'s CFO and Executive Deputy President Masanori Minamide stated in a Sep 14 Nikkei interview that rising production loads for high-end products are making it increasingly difficult to fully supply low-end products, and that the priority of defending low-end market share has effectively declined, we believe the company's fundamental policy of not changing prices based solely on short-term supply-demand conditions remains intact. That said, in our view, Murata has been gradually shifting its strategy toward focusing more on its areas of strength and adopting pricing that better reflects the need to cover large-scale capital investments, particularly in those core segments where it enjoys a competitive advantage. We believe that Murata's reluctance to enter into LTAs for AI/data-center use value added MLCCs reflects its view that locking in prices today may not be optimal, given the potential for MLCC pricing to rise in the future.

We expect sales growth to continue for small-sized, high-capacity, high-value-added MLCCs for AI servers/DCs

MLCCs are a key business, accounting for 50% of sales at Murata, 46% at SEMCO, 19% at Yageo, and 70% at Taiyo Yuden in 2025. We expect global MLCC shipment value to expand at a CAGR of 20.3%, from \$18.51bn in 2025 to \$44.45bn in 2031. We believe growth is set to accelerate from the 5.0% CAGR recorded in global MLCC shipment value over the 13 years of 2012-25 because we expect continued growth in sales of small, high-capacity, high-value-added MLCCs for AI servers/datacenters (DCs) from 2026. For AI servers/DCs alone, we forecast MLCC shipment value to reach \$23.33bn by 2031 (vs. our previous estimate of \$10.80bn), implying a 57.1% CAGR (vs. 38.1%).

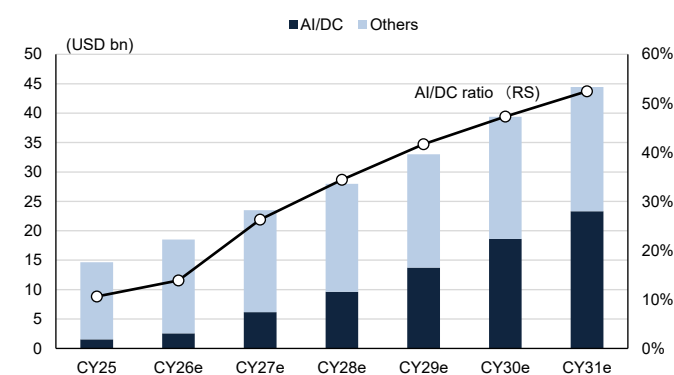
Howard Kao (our Greater China Technology Hardware analyst) expects MLCC demand volume for VR200 NVL72 racks to increase about 80% versus GB300 NVL72 racks, and forecasts the number of MLCCs mounted in each Rubin NVL72 rack will jump to roughly 570,000 units from around 320,000 for GB300 NVL72 racks, with MLCC dollar content per rack estimated to increase by 166%. We expect the rapid expansion of the high-value-added MLCC market for AI servers/DCs to be especially positive for earnings at Murata (#1 market share in MLCCs for 2025 at 41%) and SEMCO (#2 at 23%).

Exhibit 2: AI/DC-use MLCC shipment value forecasts (Old)



e=Morgan Stanley Research estimates; Source: Morgan Stanley Research

Exhibit 3: AI/DC-use MLCC shipment value forecasts (New)



e=Morgan Stanley Research estimates; Source: Morgan Stanley Research

Key roles of MLCCs in AI/DC

MLCCs play a role in locally stabilizing power supply (instantaneous current supply plus noise removal) near semiconductors, such as GPUs, CPUs, and FPGAs. Since GPUs, CPUs, and FPGAs repeatedly switch on and off on a nanosecond timescale, they demand large currents instantaneously, and MLCCs meet this instantaneous current demand by immediately discharging their stored charge. As semiconductor miniaturization advances, circuit operating speed increases and drive voltage falls. As voltage decreases, even small voltage fluctuations risk causing semiconductor malfunction, so even more stable power supply voltage is required than before. Demand for capacitor capacitance is also increasing in line with AI servers' rising computing power and power consumption, and because board area and mountable space are limited, we expect demand for smaller, higher-capacity MLCCs to persist.

Sharp dips vs. share price peaks provide re-entry opportunity

Year-to-date to Sep 11, share prices of the four companies had risen 127% for Murata, 449% for SEMCO, 141% for Yageo, and 160% for Taiyo Yuden. However, from their respective June-July peaks, these have fallen 40% for Murata, 38% for SEMCO, 52% for

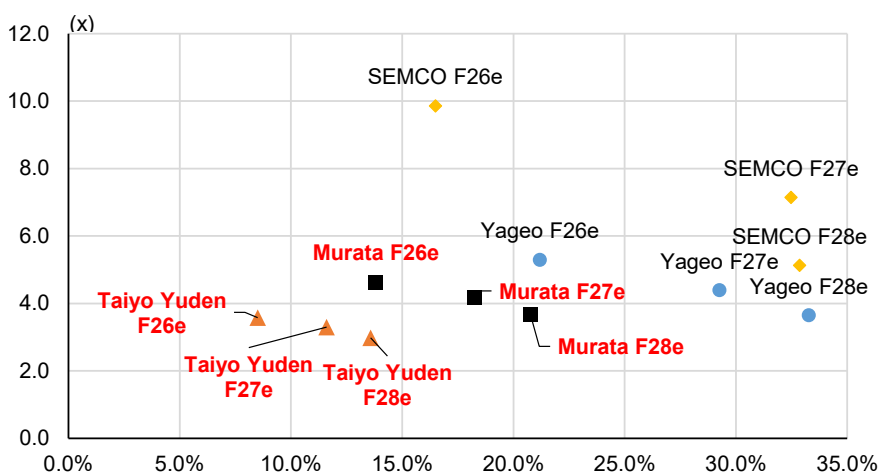
Yageo, and 59% for Taiyo Yuden. Considering the strong likelihood that high-value-added products for AI servers/DCs will continue to drive growth in MLCC demand not only in 2026 but also from 2027 onward, and that each company's earnings will stay on an expansionary trend, we believe the conditions are now in place to consider re-entering these stocks.

Exhibit 4: Stock performance comparison for 4 MLCC suppliers

	Murata	SEMCO	Yageo	Taiyo Yuden
Trading currency	JPY	KRW	NTD	JPY
Latest close (9/11)	7,352	1,400,000	544	9,209
CY26 high	12,250	2,270,000	1,140	22,655
Price target	11,000	2,620,000	1,050	10,500
YTD	126.5%	449.0%	141.2%	160.1%
Deviation from peak price	-40.0%	-38.3%	-52.3%	-59.4%
Deviation from PT	49.6%	87.1%	93.0%	14.0%

Source: FactSet, Morgan Stanley Research

Exhibit 5: ROE-P/B comparison for 4 MLCC suppliers



e=Morgan Stanley Research estimates; Source: Company data, Morgan Stanley Research

Exhibit 6: Valuation/earnings forecasts comparisons for 4 MLCC suppliers

	Murata 6981.T	SEMCO 009150.KS	Yageo 2327.TW	Taiyo Yuden 6976.T
Rating	Overweight	Overweight	Overweight	Equal-weight
Trading Currency	JPY	KRW	TWD	JPY
Price Target	11,000	2,620,000	1,050	10,500
Current Price	7,352	1,400,000	544	9,209
Upside/(Downside) (%)	50%	87%	93%	14%
Market Cap (in USD mm)	87,168.4	78,844.1	34,344.9	8,256.9
Avg Daily Traded Vol (in USD mm)	541.9	457.2	615.6	452.0
Bull Case Value	14,000	3,000,000	1,490	14,000
Upside (%)	90%	114%	174%	52%
Bear Case Value	3,500	1,350,000	480	4,500
Downside (%)	-52%	-4%	-12%	-51%
Risk/Reward Skew	1.7	32.0	14.8	1.0
Earnings Forecasts (in USD mn)				
FY26e				
Sales	14,057.9	10,258.7	5,797.2	2,746.3
MLCC Sales	8,163.7	5,374.1	1,296.3	2,055.2
(% of sales)	58.1%	52.4%	22.4%	74.8%
OP	3,143.6	1,564.6	1,589.4	288.8
Net income	2,454.8	1,219.0	1,289.4	192.6
FY27e				
Sales	16,957.0	17,387.4	7,984.2	3,135.2
MLCC Sales	10,173.9	10,988.2	2,531.9	2,414.6
(% of sales)	60.0%	63.2%	31.7%	77.0%
OP	4,533.0	4,253.2	2,663.3	426.4
Net income	3,480.9	3,340.2	2,093.2	282.9
Implied Multiples on MS Price Target				
FY26e				
P/E	52.0x	116.9x	47.2x	48.1x
EV/EBIT	39.62	86.67	28.22	33.13
EV/EBITDA	28.96	55.34	27.12	15.71
EV/Sales	8.86	13.22	5.45	3.48
FY27e				
P/E	36.4x	46.2x	41.9x	33.0x
EV/EBIT	28.62	34.72	28.86	22.00
EV/EBITDA	22.53	27.84	27.81	12.31
EV/Sales	7.65	8.49	5.46	2.99
Stock Price Performance				
1 Month	(2.5%)	10.9%	(11.8%)	(10.3%)
3 Month	(18.0%)	(22.4%)	(35.4%)	(44.6%)
1 Year	195.0%	644.7%	294.2%	203.1%
YTD	126.5%	449.0%	135.5%	160.1%

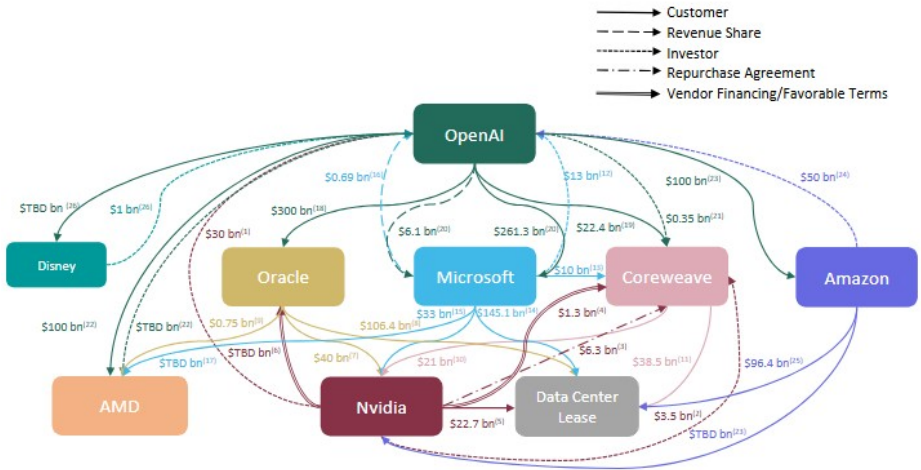
Note: e=Morgan Stanley Research estimates; Rating data is based on FactSet; FY ends in Mar for Murata and Taiyo Yuden, ends in Dec for SEMCO and Yageo; Based on Sep 11 close price
Source: FactSet, Morgan Stanley Research

Earnings and share price risks

We expect capital expenditure on AI/DC to keep expanding and generational transitions to

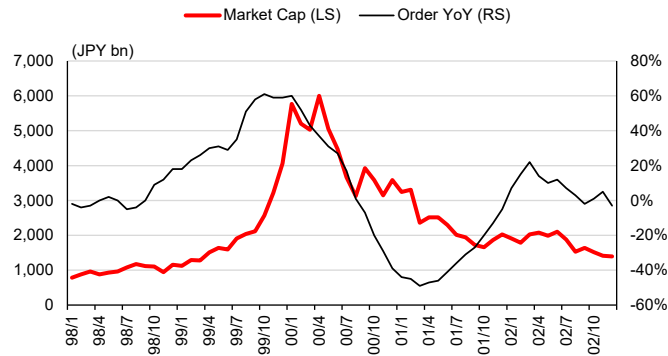
continue in GPUs, sustaining growth in demand for small, high-capacity, high-value-added MLCCs. Risks to the continuity of AI/DC capex expansion include: (1) rising risk from the growing practice of hyperscalers and major semiconductor companies charging usage fees in exchange for equity investments in AI development companies, conditional on the use of their own cloud services or semiconductors; and (2) risk that AI/DC capex growth slows due to bottlenecks, such as power infrastructure constraints. If the pace of AI/DC capex growth slows, we cannot rule out that MLCC demand falls short of our base case outlook for demand. Also, as shown in Exhibits Exhibit 8 Exhibit 8 and Exhibit 9 , during the IT bubble around 2000, there was a strong correlation between the YoY change in Murata's and Taiyo Yuden's monthly order intake and their market cap. Since order intake has risen sharply since Jul-Sep 2025, we cannot ignore the possibility that YoY growth in order intake will slow. On the other hand, MLCC demand continues to expand for high-value-added products for AI/DC, and we expect the major MLCC makers' sales to grow along with an improving product mix.

Exhibit 7: AI ecosystem capital flows



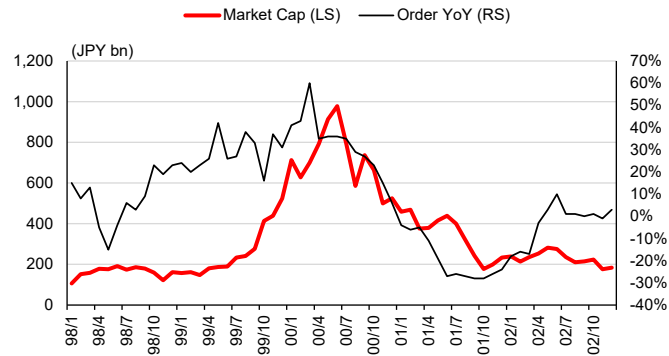
Source: Morgan Stanley Research

Exhibit 8: Murata Mfg: Market cap and order YoY (IT bubble era)



Source: Company data, Morgan Stanley Research

Exhibit 9: Taiyo Yuden: Market cap and orders YoY (IT bubble era)



Source: Company data, Morgan Stanley Research

High-value-added products for AI servers will likely drive MLCC market growth

We expect global MLCC shipment value to increase at 20.3% CAGR through 2031

As shown in [Exhibit 10](#), we estimate that global MLCC shipment value fell for two consecutive years, from \$17.47bn in 2021 (up 26.0% YoY) to \$14.32bn in 2022 (down 18.1%) and \$12.64bn in 2023 (down 11.7%), before rising back up to \$13.30bn in 2024 (up 5.2%) and \$14.67bn in 2025 (up 10.3%). We increase our forecasts for shipment value from \$18.48bn to \$18.51bn for 2026 (up 26.2% YoY), from \$22.16bn to \$24.52bn for 2027 (up 32.4%), and from \$25.73bn to \$28.60bn for 2028 (up 16.7%), and we expect the market to reach \$44.45bn in 2031, growing at a CAGR of 20.3% over the six-year period.

Exhibit 10: Global MLCC market forecasts

Base Case	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026e	2027e	2028e	2029e	2030e	2031e
Total worldwide shipment quantity (Bil pcs)	1,504	1,480	1,640	2,050	2,029	2,167	2,293	2,605	2,735	3,080	3,480	3,828	3,254	3,695	4,446	3,572	3,187	3,386	3,773	4,037	4,360	4,709	5,086	5,493	5,932
YoY (%)	14.9%	-1.6%	10.8%	24.9%	-1.0%	6.8%	5.8%	13.6%	5.0%	12.6%	13.0%	10.0%	-15.0%	13.5%	20.3%	-19.7%	-10.8%	6.2%	11.4%	7.0%	8.0%	8.0%	8.0%	8.0%	8.0%
CAGR (1991-)	14.9%	13.8%	13.7%	14.2%	13.4%	13.1%	12.8%	12.8%	12.5%	12.5%	12.4%	11.3%	11.4%	11.6%	10.5%	9.7%	9.6%	9.7%	9.6%	9.6%	9.5%	9.5%	9.4%	9.4%	9.4%
ASP in worldwide shipment (Cent)	0.47	0.42	0.36	0.38	0.39	0.36	0.33	0.30	0.29	0.26	0.27	0.37	0.39	0.38	0.39	0.40	0.40	0.39	0.39	0.43	0.48	0.52	0.54	0.55	0.57
YoY (%)	-0.2%	-10.1%	-14.0%	6.8%	1.2%	-7.7%	-7.2%	-10.6%	-4.1%	-7.2%	3.8%	34.3%	4.6%	-2.8%	4.7%	2.0%	-1.0%	-1.0%	-1.0%	11.0%	11.0%	7.5%	4.0%	3.0%	3.0%
CAGR (1991-)	-8.6%	-8.7%	-9.0%	-8.2%	-7.8%	-7.8%	-7.7%	-7.9%	-7.7%	-7.7%	-7.3%	-6.0%	-5.6%	-5.5%	-5.2%	-5.0%	-4.9%	-4.8%	-4.6%	-4.2%	-3.8%	-3.5%	-3.4%	-3.2%	-3.0%
Total worldwide shipment value (Bil \$)	7.00	6.19	5.90	7.87	7.89	7.77	7.63	7.75	7.80	8.15	9.56	14.13	12.56	13.86	17.47	14.32	12.64	13.30	14.67	17.43	20.89	24.25	27.24	30.30	33.71
YoY (%)	14.6%	-11.6%	-4.7%	33.4%	0.2%	-1.5%	-1.8%	1.6%	0.7%	4.5%	17.3%	47.7%	-11.1%	10.3%	26.0%	-18.1%	-11.7%	5.2%	10.3%	18.8%	19.9%	16.1%	12.3%	11.2%	11.2%
CAGR (1991-)	5.0%	4.0%	3.5%	4.8%	4.6%	4.3%	4.0%	3.9%	3.8%	3.8%	4.3%	5.7%	5.0%	5.2%	5.8%	5.0%	4.4%	4.4%	4.6%	5.0%	5.3%	5.6%	5.8%	5.9%	6.1%

e=Morgan Stanley Research estimates; Source: Morgan Stanley Research

AI/DC MLCC market to continue growing at a rapid pace

We expect growth in demand to accelerate for high-value-added products for AI/DC from the second half of 2026. Howard Kao expects MLCC demand volume for VR200 NVL72 racks to increase about 80% versus GB300 NVL72 racks, and forecasts the number of MLCCs mounted in each Rubin NVL72 rack will rise sharply to roughly 570,000 units from around 320,000 units for GB300 NVL72 racks ([Exhibit 11](#)). This increase stems from a rise in the number of MLCCs mounted on both compute boards and switch boards, as well as an increase in PCB content across the overall system, including modules such as CX9 Orchid. Howard Kao estimates this increase in MLCC mounting volume will lift the dollar-based MLCC content per rack by 166% ([Exhibit 12](#)).

Exhibit 11: MLCC assumptions for servers

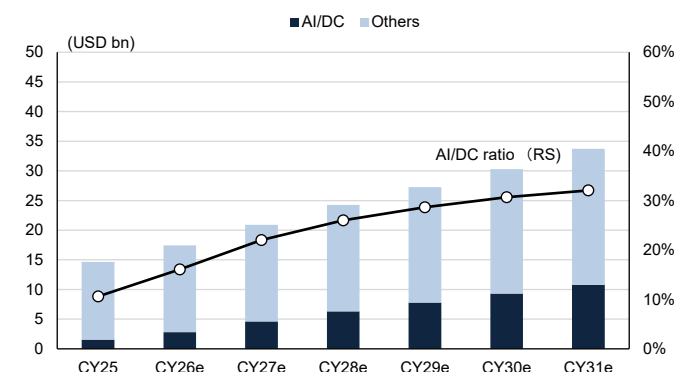
Server Assumptions		# of MLCCs		US\$	
General server		2,000		43	
HGX Hopper AI server (8x GPU)		22,000		505	
GPU module (OAM)	8x	1,300	10,400	30	240
GPU baseboard (UBB)	1x	4,900	4,900	135	135
CPU motherboard	1x	1,500	1,500	75	75
Peripheral boards	8x	650	5,200	7	55
HGX B200 AI server (8x GPU)		36,000		886	
GPU module (OAM)	8x	3,000	24,000	72	576
GPU baseboard (UBB)	1x	5,300	5,300	180	180
CPU motherboard	1x	1,500	1,500	75	75
Peripheral boards	8x	650	5,200	7	55
HGX B300 AI server (8x GPU)		48,000		1,240	
GPU module (OAM)	8x	4,400	35,200	120	960
GPU baseboard (UBB)	1x	6,100	6,100	150	150
CPU motherboard	1x	1,500	1,500	75	75
Peripheral boards	8x	650	5,200	7	55
GB200 (NVL72)		337,500		5,370	
Bianca	36x	6,800	244,800	118	4,232
Switch board	9x	3,800	34,200	57	515
Peripheral boards	90x	650	58,500	7	622
GB300 (NVL72)		319,500		4,664	
Bianca	36x	6,300	226,800	98	3,527
Switch board	9x	3,800	34,200	57	515
Peripheral boards	90x	650	58,500	7	622
VR200 (NVL72)		571,050		12,411	
Strata	36x	11,000	396,000	257	9,239
Switch board	9x	7,000	63,000	222	1,997
BlueField Module	18x	2,000	36,000	20	366
CX9 Orchid Module	72x	650	46,800	7	498
Peripheral boards	45x	650	29,250	7	311
Vera Rack		494,048		8,339	
Vera MGX	128x	3,316	424,448	56	7,131
Switch board	4x	7,000	28,000	222	888
Peripheral boards	64x	650	41,600	5	320
STX Rack		326,256		3,184	
Mainboard	16x	4,791	76,656	79	1,264
Peripheral boards	384x	650	249,600	5	1,920
LPX Rack		581,024		12,371	
LPX UBB	16x	5,374	85,984	103	1,646
LPX Module	256x	1,705	436,480	39	9,953
BF4	16x	2,000	32,000	20	326
x86 CPU Module	16x	1,660	26,560	28	446

Source: Morgan Stanley Research

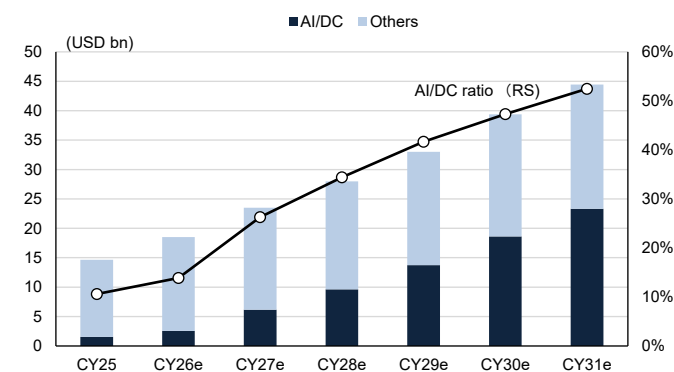
Exhibit 12: MLCC content value per rack comparison between GB300 and VR200

MLCC content per board (US\$)	GB300	VR200
Compute PCB	\$98	\$257
Switch PCB	\$57	\$222
BlueField DPU Module	\$7	\$20
ConnectX Orchid Module	\$7	\$7
Other peripheral PCB	\$7	\$7
Units per rack	GB300	VR200
Compute PCB	36x	36x
Switch PCB	9x	9x
BlueField DPU Module	0x	18x
ConnectX Orchid Module	0x	72x
Other peripheral PCB	90x	45x
Total MLCC Content per rack	GB300	VR200
Compute PCB	\$3,527	\$9,239
Switch PCB	\$515	\$1,997
BlueField DPU Module	\$0	\$366
ConnectX Orchid Module	\$0	\$498
Other peripheral PCB	\$622	\$311
Total MLCC Content per rack	\$4,664	\$12,411

Source: Morgan Stanley Research

Exhibit 13: AI/DC-use MLCC shipment value forecasts (Old)

e=Morgan Stanley Research estimates; Source: Morgan Stanley Research

Exhibit 14: AI/DC-use MLCC shipment value forecasts (New)

e=Morgan Stanley Research estimates; Source: Morgan Stanley Research

Role of MLCCs in AI servers

Capacitors, including multi-layer ceramic capacitors (MLCCs), function as a “dam” that temporarily stores electricity, and are an essential electronic component for noise removal and voltage stabilization in virtually all electronic circuits, including those used in computers, smartphones, and automobiles. MLCCs play a role in locally stabilizing power supply (instantaneous current supply plus noise removal) near semiconductors such as GPUs, CPUs, and FPGAs. Since GPUs, CPUs, and FPGAs repeatedly switch on and off on a nanosecond timescale, they demand large currents instantaneously, and MLCCs meet this instantaneous current demand by immediately discharging their stored charge. At the same time, MLCCs bypass only the AC component (noise and fluctuating current), which also serves to prevent noise from entering GPUs, CPUs, and FPGAs. MLCCs feature low ESR (low resistance) and low ESL (low inductance), allowing them to track even

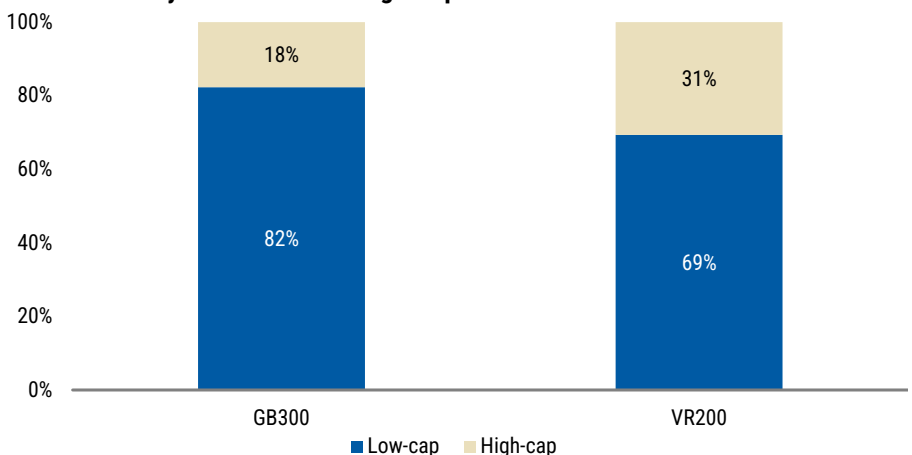
nanosecond-level current fluctuations in semiconductors such as GPUs, CPUs, and FPGAs. MLCCs are often placed around components such as GPUs and ASICs that operate at low voltages of around 1.0V. As semiconductor miniaturization advances, circuit operating speed increases and drive voltage falls. As voltage decreases, even small voltage fluctuations risk causing semiconductor malfunction, requiring an even more stable power supply voltage than before. Demand for capacitor capacitance will also rise along with AI servers' improving computing power and increasing power consumption, and because board area and mountable space are limited, demand for smaller, higher-capacity MLCCs will likely continue.

MLCC miniaturization and higher capacity to accelerate with VR200 NVL72

We assume that the space available for mounting MLCCs on VR200 NVL72 will not change significantly from the GB300 NVL72 rack, and that the transition from GB300 to VR200 will require installing higher-capacity MLCCs within the limited space of semiconductor ABF package substrates and AI accelerator boards, accelerating growth in demand for small-sized, high-capacity MLCCs. Murata has consistently been a leader in MLCC miniaturization and capacity increases, beginning mass production of 1005-size 1 F MLCCs in 2000 and 0402-size 1 F and 1005-size 22 F MLCCs in 2020. We expect demand for 1608-size 100 F, 1005-size 47 F, and 0603-size 10 F MLCCs to increase substantially with VR200 NVL72. We estimate that 1608-size 100 F MLCCs command a unit price 5-10x higher than 2012-size 100 F or 1608-size 47 F MLCCs, and we expect the shift to smaller, higher-capacity MLCCs accompanying the transition to VR200 NVL72 to drive a substantial rise in ASP.

Exhibit 15: High-cap (47uF+) MLCC usage to increase to 30%+ in VR200 (vs <20% for GB300)

Nvidia's Rubin System Uses More High-Cap MLCCs



Source: Morgan Stanley Asia Pacific Research estimates based on supply chain checks. Note: High-cap refers to 47uF+ MLCCs.

High-capacitance MLCCs (47uF+) are likely to capture a disproportionate share of this incremental opportunity: Our analysis suggests that 47uF+ MLCCs could account for approximately 50-60% of the incremental Cloud AI MLCC TAM in CY27, representing an additional ~US\$2B market opportunity alone. This reflects both the increasing use of high-

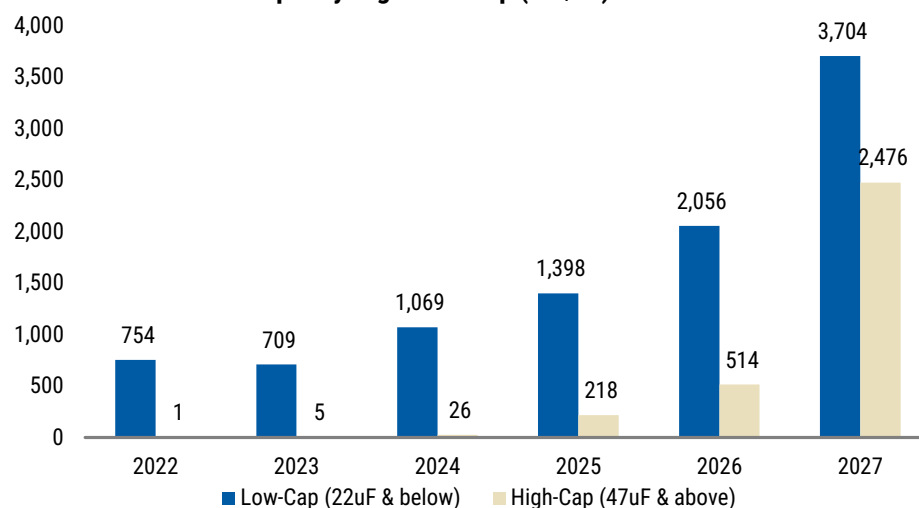
capacitance devices within AI accelerator systems and a growing mix shift toward higher-value MLCC content.

From a volume perspective, we estimate Cloud AI demand for 47µF+ MLCCs could increase from approximately 4 billion units in CY25 to 40B+ units by CY27: Such a rapid increase would represent a meaningful step-up in industry demand and could require substantial capacity reallocation or expansion among leading suppliers.

Based on our industry discussions, the market for high-capacitance MLCCs remains concentrated among a limited number of suppliers, primarily Japanese and Korean manufacturers, including Murata, Taiyo Yuden, and Samsung Electro-Mechanics (SEMCO). As demand from AI infrastructure accelerates, we believe capacity utilization for these products could tighten, raising the possibility of capacity crowding-out effects across other end markets and applications.

Exhibit 16: Demand for high-cap (47uF+) MLCC is growing at a very fast pace

AI/DC MLCC Demand Split by High/Low-Cap (US\$ M)



Source: Morgan Stanley Research; estimates are by Morgan Stanley Research

Exhibit 17: MLCC maximum capacitance by size

(µF)							
<=1000	220						
<=100		100	100	47			
<=10					10		
<=1						1	0.1
(mm)	3216	2012	1608	1005	0603	0402	0201
Murata Mfg	○	○	○	○	○	○	○

Note: ○ denotes products under mass production

Source: Morgan Stanley Research based on Murata Mfg product catalog

Exhibit 18: MLCC size comparison

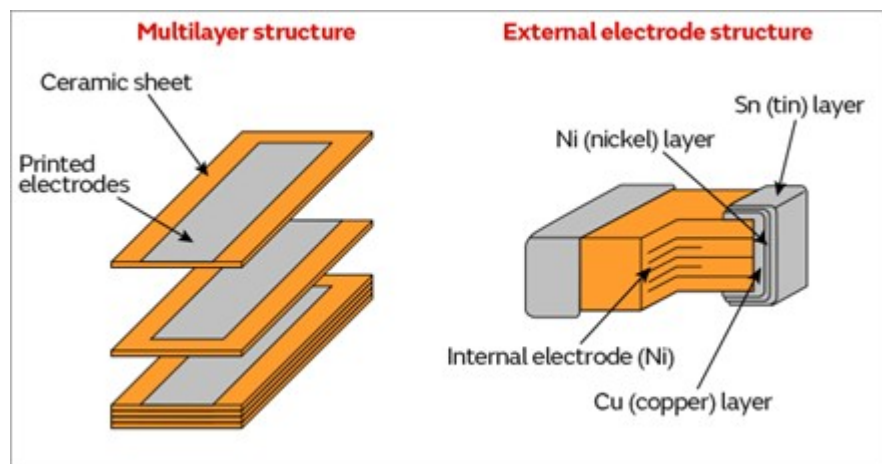
	4532	3216	2012	1608	1005	0603	0402	0201	016008
Length	4.5mm	3.2mm	2.0mm	1.6mm	1.0mm	0.6mm	0.4mm	0.25mm	0.16mm
Width	3.2mm	1.6mm	1.25mm	0.8mm	0.5mm	0.3mm	0.2mm	0.125mm	0.08mm
Thickness	2.0mm	0.85mm	0.6mm	0.8mm	0.5mm	0.3mm	0.2mm	0.125mm	0.08mm
Volume	28.8mm ³	4.35mm ³	1.5mm ³	1.024mm ³	0.25mm ³	0.054mm ³	0.016mm ³	0.004mm ³	0.001mm ³
Volume ratio	7,373	1,114	384	262	64.0	13.8	4.1	1.0	0.26

Source: Morgan Stanley Research

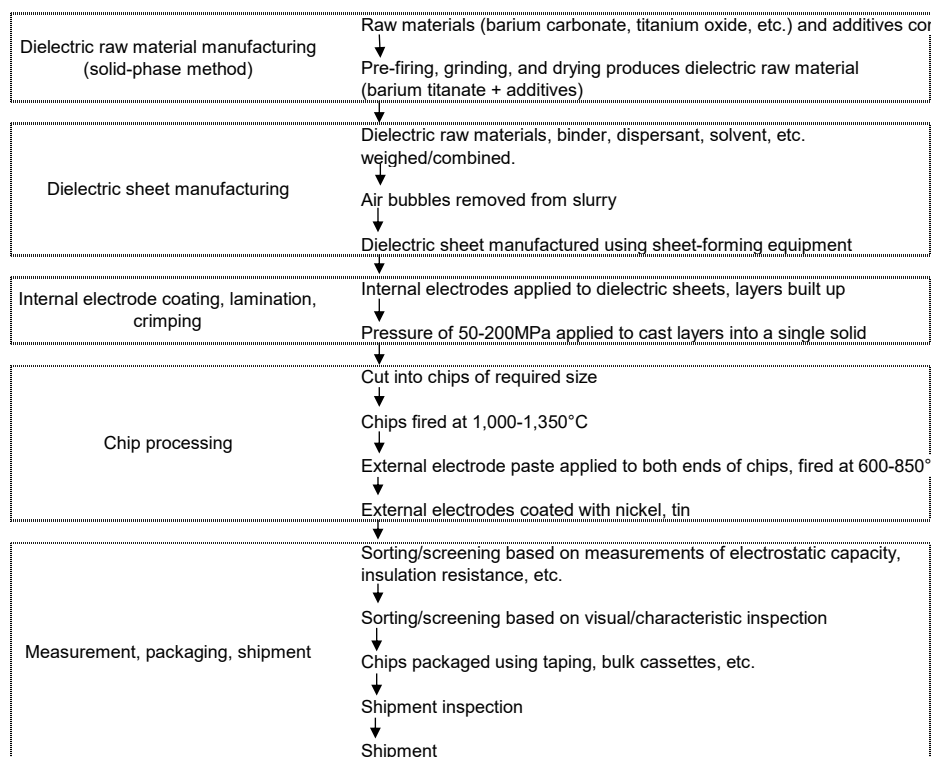
Materials development and production technology behind MLCC miniaturization and capacity increases

In MLCCs, smaller sizes and higher capacities have mainly been achieved by thinning the dielectric and electrode layers (layer thinning) and increasing the number of layers stacked (multilayering). For high-capacity products for AI/DC, in addition to the layer thinning and multilayering described above, efforts are accelerating to further miniaturize and increase capacity by expanding the area of the dielectric and internal electrodes per dielectric sheet. We believe MLCC miniaturization and capacity increases depend on: (1) developing superior dielectric and electrode materials; (2) manufacturing highly uniform, highly reliable slurry; (3) technology for precisely stacking dielectric sheets; and (4) technology for firing in a controlled atmosphere without oxidation. For high-capacity products for AI/DC, we see the key factors as (1) materials development that increases the crystallinity of dielectric materials while making them finer, in order to thin each dielectric layer as much as possible (layer thinning), together with (2) manufacturing technology to precisely stack the dielectric sheets.

Exhibit 19: MLCC composition



Source: Murata Mfg materials

Exhibit 20: MLCC production process

Source: Morgan Stanley Research

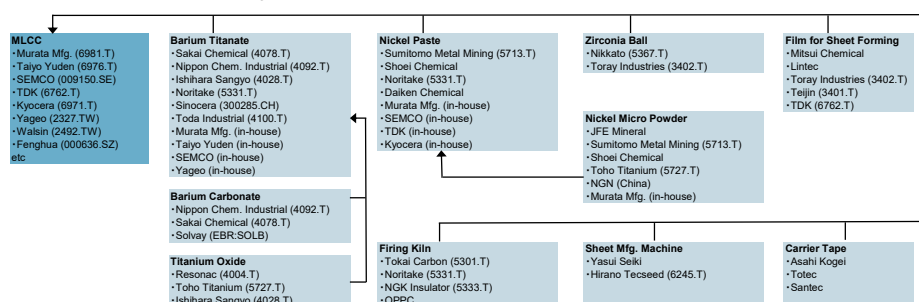
Dielectric material manufacturing process

Dielectric materials are manufactured by adding a binder, solvent, and additives to barium titanate (BaTiO_3), which is produced from barium carbonate (BaCO_3) and titanium oxide (TiO_2). Methods for manufacturing barium titanate are broadly divided into the solid-phase method and the liquid-phase method. In the solid-phase method, barium carbonate and titanium oxide are mixed and fired at high temperature, producing barium titanate through a chemical reaction at the firing stage. In the liquid-phase method, barium carbonate and titanium oxide are dissolved in a solution, reacted at the solution stage, and fired at a relatively low temperature to produce the compound. Manufacturing methods used in mass production via the liquid-phase method include the oxalate method and the hydrothermal synthesis method. Barium titanate with small and uniform particle size is key to smaller and higher-capacity MLCCs. In 2023, Murata established MF Materials, a joint venture with Ishihara Sangyo and Ishihara Sangyo's wholly owned subsidiary Fuji Titanium (with equity stakes of 20% for Murata, 10% for Ishihara Sangyo, and 70% for Fuji Titanium). MF Materials produces barium titanate via the oxalate process and supplies it to Murata Manufacturing. MF Materials is constructing a new building adjacent to its head office plant in Nobeoka, Miyazaki Prefecture, with operations scheduled to commence in 2027. TDK established a joint venture with Nippon Chemical Industrial in April 2026 called TDK-NCI Advanced Materials (with equity stakes of 51% for TDK and 49% for Nippon Chemical Industrial), with plans to accelerate development of the oxalate method barium titanate.

Exhibit 21: Situations around Barium Titanate production

	Solid-phase method	Oxalate method	Hydrothermal synthesis
Internal production			
Murata Mfg.	○		
Taiyo Yuden	○		
SEMCO	○		
Yageo	○		
External sourcing			
Sakai Chemical			○
Nippon Chem. Industrial		○	
Ishihara Chemical		○	
Noritake	○		
Toda Kogyo			○
Sinocera	○		○

Source: Morgan Stanley Research

Exhibit 22: MLCC supply chain

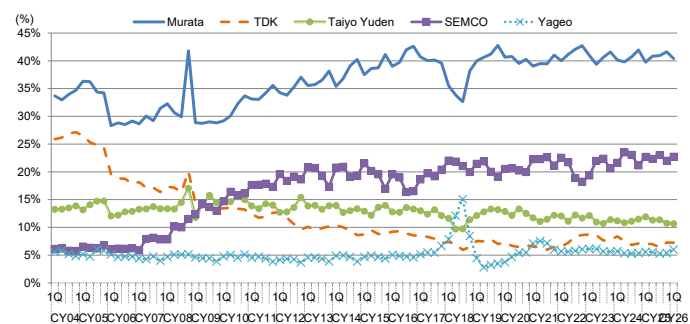
Source: Morgan Stanley Research

Murata and SEMCO to benefit the most from growing demand for high-value-added products for AI servers

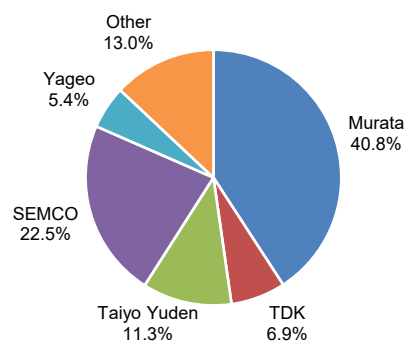
We expect Murata and SEMCO, which supply small, high-capacity products stably and in large volumes, to capture the greatest benefits from growing demand for small-sized, high-capacity MLCCs used in AI servers and data centers. Murata has explained that AI/DC MLCC sales accounted for 10-15% of its MLCC sales in F3/26 and are forecast to rise 100-105% YoY in F3/27, accounting for 20-25% of MLCC sales. We estimate Murata's AI/DC MLCC sales to rise roughly 120% YoY in F3/27 and about 100% YoY in F3/28, with AI/DC MLCCs underpinning growth in the company's overall MLCC sales. Murata has stated that it forecasts a 100-105% increase in AI/DC MLCC sales in F3/27, reflecting a roughly 40% increase in volume and a roughly 50% rise in ASP, but we expect the increase in ASP to exceed its forecast.

2025 MLCC market share was Murata 40.8%, Taiyo Yuden 11.3%

We estimate that global MLCC shipment value totaled \$14.67bn in 2025 (up 10.3% YoY), with shares of 40.8% for Murata, 22.5% for SEMCO, 11.3% for Taiyo Yuden, 6.9% for TDK, and 5.4% for Yageo, and a combined 13.0% for all companies outside these top five. We expect demand for high-value-added products for AI/DC to keep expanding from 2027, with Murata and SEMCO continuing to gain market share.

Exhibit 23: MLCC market share

Source: Company data, Morgan Stanley Research

Exhibit 24: MLCC market share (CY25)

Source: Company data, Morgan Stanley Research

What are the risks to Murata?

Risks for Murata include: (1) losing MLCC market share by failing to adequately supply MLCCs for AI servers/DCs amid rapidly expanding demand. (2) Murata's products, including MLCCs, are used not only in AI servers/DCs but across virtually all electronic equipment, including automobiles, smartphones, and industrial machinery, making the company sensitive to macroeconomic conditions and inventory cycles. (3) The company has explained that a ¥1/\$ appreciation of the yen would reduce operating profit by ¥4.5bn per year, and the yen could appreciate beyond our assumed rate of ¥155/\$, among other factors.

Murata Manufacturing earnings forecasts

We expect Murata at 2Q results (end-Oct) to hike F3/27 guidance; continued earnings growth from F3/28

F3/27-28 and F3/27 2Q forecasts: We project F3/27 OP of ¥490.8bn (guidance ¥430bn, FactSet consensus ¥453.9bn), up ¥208.9bn YoY and above F3/22's record of ¥424.1bn. That is because we expect profit in the Components segment to rise ¥186.2bn to ¥501.3bn and losses in the Device & Modules segment to shrink by ¥24.1bn from -¥26.5bn to -¥2.4bn. We look for Components profit to rise due to a +31.6% YoY increase in MLCC sales in dollar terms (+36.1% on yen basis), a continued increase after F3/26's +13.8% rise (+12.6% on yen basis), product mix improvement, and higher utilization rates absorbing increased depreciation and other fixed cost growth. We expect Device & Modules losses to shrink mainly due to the falling out of ¥43.8bn in Resonant goodwill impairment and ¥6bn from the disposal of the micro battery business. We project F3/28 OP of ¥702.6bn (consensus ¥655.3bn). The projected ¥211.8bn YoY growth breaks down as Components +¥170bn and Devices & Modules +¥39.6bn. We expect Components profit to rise thanks to a 24.5% YoY increase in capacitor (MLCC) sales in dollars (+23.7% on yen basis) and continued product mix improvement absorbing increased depreciation & other fixed cost growth. We look for Devices & Modules profit to rise on factors including increased sales of high-frequency modules, secondary batteries, and vertical power delivery modules.

We project 2Q OP of ¥125bn (consensus ¥122.9bn). The projected QoQ ¥26.6bn growth breaks down as Components +¥12.5bn QoQ to ¥126bn, and Devices & Modules +¥13.9bn QoQ to ¥1bn. For Components, we expect sales of MLCCs for AI/data centers to continue to increase, the MLCC product mix to improve, and utilization rates to continue to rise. For Devices & Modules, we expect sales for North American smartphones to increase seasonally and reenter the black, but because North American smartphone production is split between fall models and spring models, profit contributions are likely to be limited.

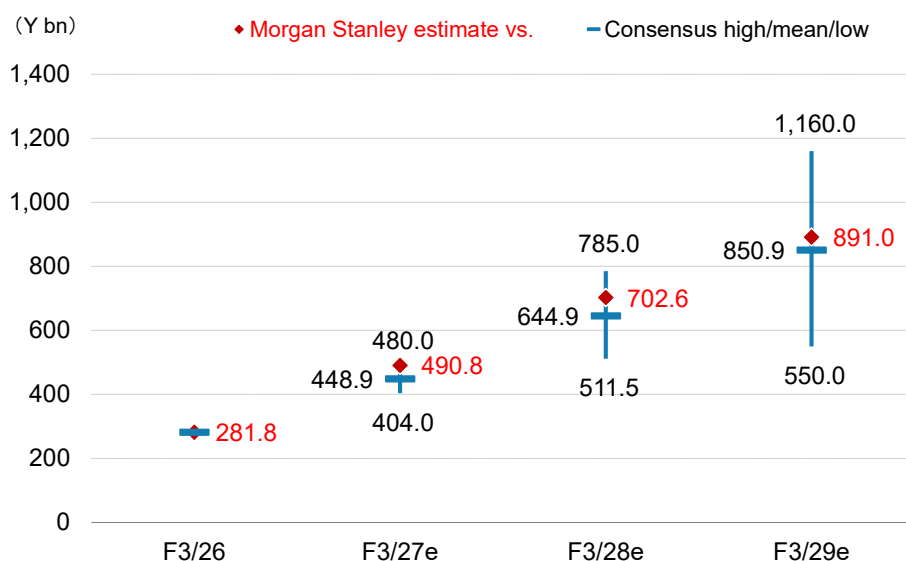
Exhibit 25: Murata Mfg: Earnings forecast changes

(Y bn)	F3/26								F3/27				F3/26				F3/27				F3/25				F3/26				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/30e				F3/31e				F3/27e				F3/28e				F3/29e				F3/			
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Exhibit 26: Murata Mfg: Mid-term forecasts

(bn JPY, JPY/share)	F3/22	F3/23	F3/24	F3/25	F3/26	F3/27 MSe	F3/28 MSe	F3/29 MSe	F3/30 MSe	F3/31 MSe
Sales	1,811.2	1,686.8	1,640.2	1,743.4	1,830.9	2,194.6	2,628.3	3,104.1	3,682.2	4,348.3
Components	984.3	911.2	933.8	1,033.1	1,159.7	1,526.1	1,838.7	2,156.5	2,528.1	2,918.2
Capacitors	788.5	738.8	753.5	831.8	936.4	1,274.5	1,577.0	1,884.3	2,245.0	2,623.8
Inductors and EMI Filters	195.8	172.3	180.3	201.3	223.3	251.6	261.7	272.2	283.1	294.4
Devices/Modules	815.0	761.0	695.2	697.2	656.0	653.2	776.7	934.7	1,141.1	1,417.0
High Frequency Devices/Communications Module	528.2	453.6	440.1	443.6	394.8	377.9	453.5	544.2	653.0	783.6
Battery/Power supply	180.4	214.6	164.4	155.7	154.1	162.3	203.4	263.5	353.5	490.7
Functional Devices	106.4	92.8	90.7	97.8	107.1	113.0	119.8	127.0	134.6	142.7
Others	11.9	14.7	11.2	13.1	15.2	15.3	13.0	13.0	13.0	13.0
OP	424.1	297.9	215.4	279.7	281.8	490.8	702.6	891.0	1,112.0	1,350.6
Components	355.5	280.1	234.2	275.2	315.1	501.4	671.4	834.5	1,029.9	1,236.3
Devices/Modules	69.7	20.6	-13.0	10.0	-26.5	-2.4	37.2	62.5	88.0	120.3
Others	-1.2	-2.8	-5.7	-5.4	-6.8	-8.2	-6.0	-6.0	-6.0	-6.0
OPM	23.4%	17.7%	13.1%	16.0%	15.4%	22.4%	26.7%	28.7%	30.2%	31.1%
ROE	15.0%	10.9%	7.3%	9.1%	8.8%	13.8%	18.2%	20.8%	22.6%	23.4%
EPS	163.7	133.8	95.7	125.1	127.7	211.5	302.0	385.3	484.1	592.5
FCF	209.2	118.4	288.1	243.8	231.4	208.5	341.2	468.1	580.4	731.8

MSe=Morgan Stanley Research estimates; Source: Company data, Morgan Stanley Research

Exhibit 27: Murata Mfg: OP forecasts comparison

Note: Consensus is based on FactSet; Source: Fact Set, Morgan Stanley Research

Murata's share price risk/reward: Even though we raise our earnings estimates for Murata, we lower our DCF-based price target from ¥12,500 to ¥11,000, mainly reflecting an increase in our DCF model's risk-free rate from 2.60% to 3.0%. Murata's share price has fallen 41% year-to-date from its June 22 peak, and we see 53% upside over the next 12-18 months, maintain our OW rating, and designate the stock as our Top Pick in Japan's electronic components industry. The reasons are as follows: (1) We forecast the dollar-based shipment value of MLCCs, where Murata held the top global share of 40% in 2025, to expand at a CAGR of 21% through 2031, with Murata continuing to gradually gain market share. (2) Demand for MLCCs used in AI servers/DCs is likely to continue expanding, and demand for small, high-capacity, high-value-added products, which Murata has brought to market ahead of competitors, should grow substantially. (3) Outside of MLCCs, we look for demand to expand for small, high-value-added products, including secondary batteries for DC BBUs, power modules for vertical power delivery in AI servers, and RF modules for smartphones, and then expand even further as edge AI devices

become widespread. (4) In addition to earnings growth, we expect that management's proactive approach to shareholder returns, including share buybacks, will raise capital efficiency and lift ROE.

Bull case ¥14,000, 95% projected upside

In our bull case, we assume higher sales of MLCCs for AI servers/data centers than in our base case, for component sales of 7% higher than our base case, for OP of ¥560.3bn in F3/27, ¥794bn in F3/28, and ¥1.515tn in F3/31. Our forex assumptions and DCF assumptions are the same as for our base case. Bull case fair value corresponds to 58.2x of fully-diluted F3/27 EPS of ¥240.4, 41.1x fully-diluted F3/28 EPS of ¥340.7, and 21.1x fully-diluted F3/31 EPS of ¥664.0.

Bear case ¥3,500

Murata products are used in all kinds of different electronic devices, and they are easily affected by the macro economy and inventory cycle. In our bear case, we assume slower global economic growth rates and lower production of high-end consumer goods like high-end smartphones than in our base case for OP of ¥314.6bn in F3/27, ¥330bn in F3/28, and ¥362.4bn in F3/31. Our forex assumptions and DCF assumptions are the same as in our base case. Here we look for ROE to increase from 8.8% in F3/26 to 9.3% in F3/27 and 9.4% in F3/28, above capital cost of 7.5%. With profit below the market consensus, we assume the share price falls to a P/B of 2.4x.

Risk Reward – Murata Manufacturing (6981.T) Top Pick

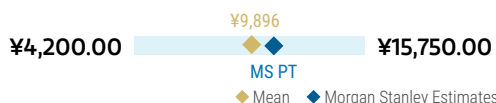
We expect AI/DC use value-added MLCC sales and earnings to continue to expand

PRICE TARGET ¥11,000

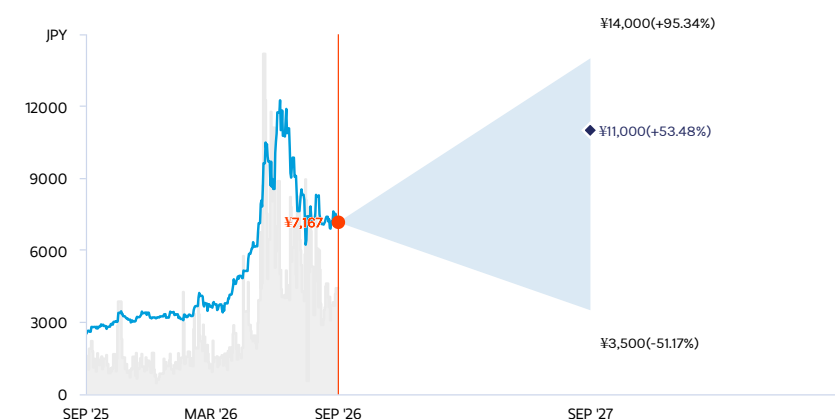
Derived from our base case and a DCF model, with the following assumptions: 3.0% risk-free rate, 1.39 equity beta, 3.2% risk premium, yielding 7.5% WACC; zero growth from F3/36.

Consensus Price Target Distribution

Source: Refinitiv, Morgan Stanley Research



RISK REWARD CHART



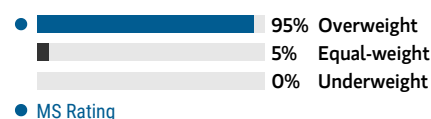
Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- We expect AI/DC use value-added MLCC sales and earnings expansion to continue..
- Expect a significant increase in the total capacitance required for MLCCs in AI/data center applications, alongside continued miniaturization and further increases in capacitance per unit.
- We expect Murata to remain competitive in MLCCs and other products over the long term.
- In F3/27 onward, we expect module earnings to improve as sales of RF modules rise.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

New Data Era: *Positive*

Secular Growth: *Positive*

View descriptions of Risk Rewards Themes [here](#)

BULL CASE

¥14,000

Derived from DCF

Demand for AI/DC use MLCC is stronger than in the base case. OP assumptions: ¥560.3bn in F3/27, ¥794.0bn in F3/28, ¥1,515bn in F3/31. Same FX & DCF model assumptions as the base case. Bull case fair value equates to P/E of 58.2x on F3/27 diluted EPS of ¥240.4, 41.1x on F3/28 diluted EPS of ¥340.7, and 21.1x on F3/31 diluted EPS of ¥664.0.

BASE CASE

¥11,000

Derived from DCF

Earnings continue to grow and beat consensus over the medium/long term due to technological advantages & competitiveness, especially in MLCCs. Assumptions: OP ¥490.8bn in F3/27, ¥702.6bn in F3/28, ¥1,350.6bn in F3/31; FX ¥155/US\$, DCF model (see Valuation Methodology); zero growth from F3/36. Base case fair value equates to P/E 52.0x on F3/27e diluted EPS of ¥211.5, 36.4x on F3/28e diluted EPS of ¥302.0 and 18.6x on F3/31e diluted EPS of ¥592.5.

BEAR CASE

¥3,500

F3/27e BPS ¥1,486 x P/B 2.4

Murata remains competitive, but final demand for its components is lower than we envision. OP assumptions: ¥314.6bn in F3/27, ¥330.0bn in F3/28, ¥362.4bn in F3/31. Same FX assumptions as the base case. Assumes ROE moves from 8.8% in F3/26 to 9.3% in F3/27 and 9.4% in F3/28, higher than our capital cost estimate of 7.5%, but with earnings missing market expectations, P/B drops to 2.4x.

Risk Reward – Murata Manufacturing (6981.T)

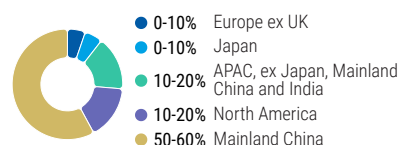
KEY EARNINGS INPUTS

Drivers	Mar 2026	Mar 2027e	Mar 2028e	Mar 2029e
Components Total OPM (%)	27.2	32.9	36.5	38.7
Capex (¥, bn)	247.8	253.0	250.0	250.0
Capacitors Revenue (¥, bn)	936.4	1,274.5	1,577.0	1,884.3

INVESTMENT DRIVERS

- Strength of global manufacturing.
- Over the medium/long term, we think Murata can maintain its tech positioning (MLCC global share of 40%) and increase earnings.

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

5/5
MOST

3 Month
Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

If total MLCC capacitance in AI/DC applications exceeds our forecast, driving sales of high value-added MLCCs (smaller, higher capacitance) above our forecasts.

RISKS TO DOWNSIDE

- Since Murata's core products are in all sorts of electronic devices, demand and unit prices could fluctuate widely due to changes in the global economy.
- Slower MLCC technology innovation.
- We estimate ¥1/\$ weakness boosts annual OP by ¥4.5bn.

OWNERSHIP POSITIONING

Inst. Owners, % Active **67.7%**

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Mar 2028e

Sales / Revenue (¥, bn) 2,160.0 **2,628.3** 2,735.0
2,462.7

EBITDA (¥, bn) 646.0 **893** 988.0
849.3

Net income (¥, bn) 362.0 **540** 609.1
499.8

EPS (¥) 198.9 **302.0** 350.0
276.7

◆ Mean ◆ Morgan Stanley Estimates
Source: Refinitiv, Morgan Stanley Research

SEMCO earnings forecasts

Shawn Kim

AI-driven MLCC growth and pricing to drive sharp earnings acceleration in FY27, with continued growth in FY28

FY26-28 forecasts: We project SEMCO OP of W2.34tn in FY26 (+156% YoY), W5.70tn in FY27 and W8.36tn in FY28, implying further growth of 144% and 47%, respectively. We expect the earnings acceleration to be driven mainly by the Components (MLCC) and Package Solution businesses, while the Optics business remains a relatively modest contributor. In FY27, our projected OP of W5.70tn breaks down as Components W4.56tn, Package Solution W943bn, and Optics W201bn, versus W1.67tn, W510bn, and W165bn, respectively, in FY26. In FY28, we forecast Components OP rising further to W6.55tn, Package Solution to W1.58tn, and Optics to W230bn. As such, we expect SEMCO's consol. OP margin to grow from 8.1% in FY25 to 15.7% in FY26, 24.5% in FY27, and 25.9% in FY28. We maintain the stock as our top pick within our S. Korea Technology coverage.

For Components, we expect profit growth to accelerate as AI-server demand drives both strong MLCC volume growth and a structurally richer product mix. High-end AI MLCC supply remains tight, with lead times exceeding 20 weeks, supply-chain inventories lean following two years of destocking, and book-to-bill at the three high-end suppliers reaching its highest level since the pandemic. Importantly, contract pricing has increasingly been institutionalized through long-term agreements, while the high-end AI MLCC market remains concentrated, with Murata and SEMCO together holding around 85% share. Capacity additions are also unlikely to close the supply-demand gap quickly, as most meaningful new capacity is scheduled for 2027 or later and suppliers are generally targeting only around 10-15% annual output growth. We thus expect tighter supply-demand, higher pricing, and a shift toward high-capacitance, low-ESL AI products to drive Components OP margin from 21.4% in FY26 to 30.9% in FY27 and 30.1% in FY28.

Bull case W3,000,000

Our bull case assumes ABF opportunities from major US CSPs drive stronger content growth, with future ASIC content & volume growth continuing to outpace our base-case expectations. Customer diversification toward non-mobile customers should also reduce earnings volatility and improve profitability. Our bull case, equivalent to 15x FY27e P/B, reflects what we see as an unprecedented super-cycle in the underlying AI-related MLCC and ABF markets.

Bear case W1,350,000

In our bear case, we assume slower content growth in MLCCs and substrates due to greater competition and China's localization, together with prolonged consumer weakness weighing on the Optics business. However, we still expect increasing AI exposure to structurally lift SEMCO's MLCC and ABF ASPs, product mix and margins, given the difficulty commodity suppliers face in penetrating the premium market. Our bear-case value corresponds to 7x FY27e P/B. Key downside risks include a significant pullback in

Samsung's mobile flagship cycle, lower SEMCO market share, execution risk in penetrating Chinese smartphone customers, and continued consumer-demand weakness.

Risk Reward – Samsung Electro-Mechanics (009150.KS)

Top Pick

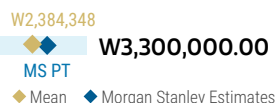
Dual AI opportunity – ABF and MLCC

PRICE TARGET W2,620,000

Base case, residual income valuation model. We apply a 10% cost of equity (beta 1.0, equity risk premium of 6.5%, and risk-free rate of 3.5%) and a 5% terminal growth rate. Our base year is 2026 and our forecast period is 2026-36.

Consensus Price Target Distribution

W570,000.00



Source: Refinitiv, Morgan Stanley Research

RISK REWARD CHART AND OPTIONS IMPLIED PROBABILITIES (12M)



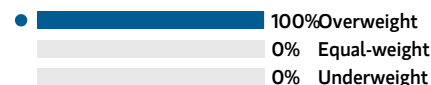
Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research, Morgan Stanley Institutional Equities Division. The probabilities of our Bull, Base, and Bear case scenarios playing out were estimated with implied volatility data from the options market as of 14 Sep 2026. All figures are approximate risk-neutral probabilities of the stock reaching beyond the scenario price in either three-months' or one-years' time. View explanation of Options Probabilities methodology [here](#)

OVERWEIGHT THESIS

- We see long-term improvement in SEMCO's operating profile – multi-layer core ABF capacity sold out from ASIC customers and accelerating AI penetration in MLCC, driving substantial content from rising complexity.
- With the AI computing opportunity and the prospect of a cyclical recovery from 2H26, earnings expectations appear to be lagging significantly based on the current outlook.
- We think the share price will look to discount accelerating EPS estimate revisions.
- Our price target implies 2027e P/B of 13x.

Consensus Rating Distribution



MS Rating

Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Secular Growth: Positive

View descriptions of Risk Rewards Themes [here](#)

BULL CASE

W3,000,000

15x 2027E P/B

ABF opportunities from major US CSPs drive meaningful content growth, and we estimate future ASIC content and volume growth will continue to outpace. Customer diversification into non-mobile companies also lowers earnings volatility and improves profitability. Our bull case implies a 2028e P/B at 15x. We believe this premium is supported by what we view as an unprecedented super-cycle in the underlying industry dynamics.

BASE CASE

W2,620,000

13x 2027e P/B

We see a number of catalysts ahead for SEMCO, including a large ABF opportunity in ASIC chip wins, accelerating MLCC growth and FCF inflection. Our long-term conviction in custom silicon opportunities via ABF substrates has grown substantially.

BEAR CASE

W1,350,000

7x 2027e P/B

We reflect slower content growth for MLCC and substrates amid competition and China's localization. But AI exposure will structurally increase SEMCO's MLCC and ABF ASP, product mix, and margin with commodity suppliers not easy to penetrate.

Risk Reward – Samsung Electro-Mechanics (009150.KS)

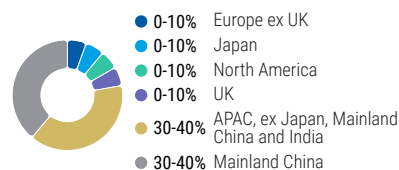
KEY EARNINGS INPUTS

Drivers	Dec 2025	Dec 2026e	Dec 2027e	Dec 2028e
Revenue Growth (%)	9.9	31.5	56.7	38.6
OPM (%)	8.1	15.7	24.5	25.9
CAPEX (%)	8.7	14.3	9.8	10.4

INVESTMENT DRIVERS

- AI servers, high-speed networking, Auto 2.0, and smartphone content growth
- Customer diversification away from Samsung Electronics, targeting major CSPs
- MLCC pricing/margin trend

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

3/5
MOST

3 Month
Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Price hikes driven by MLCC shortage
- Better-than-expected smartphone demand
- Higher MLCC margin from better yield and product mix
- Strong tailwind from China's consumption-friendly policies

RISKS TO DOWNSIDE

- Significant pullback in Samsung's mobile flagship product cycle and SEMCO's lower market share
- Execution risk in penetrating Chinese smartphone customers
- Consumer demand headwinds in 2026

OWNERSHIP POSITIONING

Inst. Owners, % Active **84.5%**

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e

Sales / Revenue (W, bn) 12,696 **14,875**
14,217

EBITDA (W, bn) 2,246 **3,553**
2,965

EBIT (W, bn) 1,256 **2,269**
1,909

EPS (W) 12,718 **22,407**
19,809

◆ Mean ◆ Morgan Stanley Estimates
Source: Refinitiv, Morgan Stanley Research

Yageo earnings forecasts

Howard Kao

Tightening MLCC Narrative is Unchanged

We forecast CY26 revenue of approximately NT\$183B, up 38% y/y, driven primarily by stronger AI-related demand, with a more modest contribution from consumer applications. This demand strength should support broad-based utilization improvement across Yageo's passive component portfolio. Pricing is also turning more favorable, with distributor price increases implemented in 2Q26 and Yageo raising prices for direct customers in 3Q26, [as previously highlighted](#). The combination of higher utilization and firmer pricing should drive meaningful operating leverage. We forecast gross margin expanding to 39.3%, up 310bps y/y, and operating margin reaching 27.4%, up 500bps y/y, supporting EPS of approximately NT\$19.79.

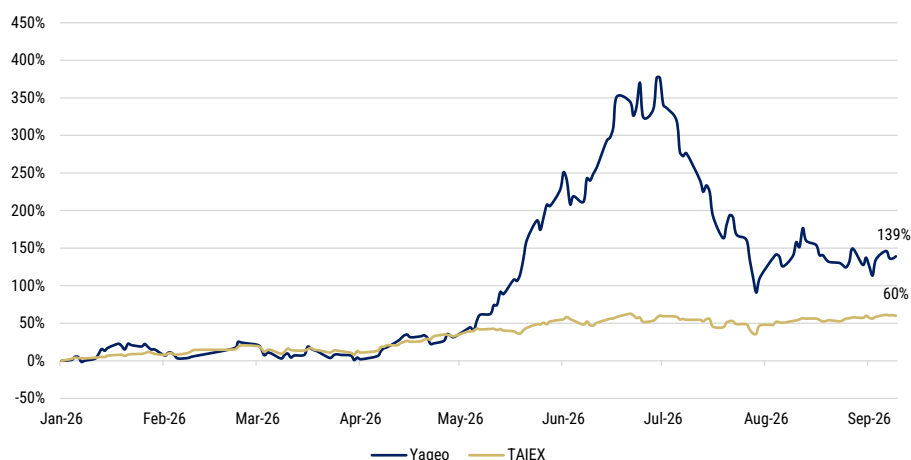
Looking into CY27, we expect supply/demand conditions in the mid- and low-end markets to tighten further, sustaining pricing momentum and driving an additional improvement in utilization. We therefore forecast revenue of approximately NT\$252B, up 38% y/y. Continued operating leverage should lift gross margin to 43.2%, an increase of 390bps y/y, and operating margin to 33.4%, up 600bps y/y, resulting in EPS of approximately NT\$32.11.

Why has the stock been so volatile?

Yageo benefited from a sharp increase in MLCC pricing in 2Q26, driven by a temporary spike in spot-market demand. In our view, ODM/EMS customers accelerated inventory replenishment amid concerns that the ramp in high-end AI-server MLCC demand, particularly alongside Nvidia's Vera Rubin platform, could constrain supply of mid- and low-end MLCCs. Under normal market conditions, these customers typically carry only 3-4 weeks of inventory. However, supply concerns prompted a rapid increase in inventory coverage to approximately 4-6 months.

During this restocking phase, ODM/EMS customers sourced MLCCs from both manufacturers, including Yageo, and distributors. The resulting demand surge drove a sharp increase in spot/distributor pricing and also contributed to the stock's strong performance. However, the pricing move was largely driven by a one-off inventory build. Once customers had secured sufficient inventory, purchasing activity normalized and spot pricing began to soften. This, together with weaker broader market sentiment, led to a pullback of ~50% from the stock's peak of approximately NT\$1,140.

At the current share price, however, we believe the risk-reward has become attractive. The underlying fundamentals, and, importantly, our view of the MLCC upcycle, remain intact. Our analysis indicates substantial incremental demand for high-end, high-capacitance MLCCs of 47µF and above, which will require a meaningful increase in dedicated production capacity. We expect this to drive further capacity conversion toward high-end MLCCs, with the resulting crowding-out effect becoming more visible from late CY26 into early CY27. This should progressively tighten supply in the mid- and low-end MLCC market and provide renewed support for pricing.

Exhibit 28: Yageo vs TAIEX YTD share price performance (%)**Yageo vs. TAIEX YTD performance**

Source: Company data, Morgan Stanley Research.

Earnings Estimates Change

We have fine-tune our model, which drives our CY26-28 EPS reductions by 2%, 6%, and 7% respectively.

Exhibit 29: Yageo: Earnings Estimates Change

NT\$ mn	2026e Earnings Estimates			2027e Earnings Estimates			2028e Earnings Estimates		
	New	Old	Variance	New	Old	Variance	New	Old	Variance
Net sales	182,769	182,769	0%	251,501	251,501	0%	306,659	306,659	0%
COGS	-110,968	-110,474	0%	-142,878	-138,732	3%	-165,373	-157,970	5%
Gross profit	71,801	72,295	-1%	108,623	112,769	-4%	141,286	148,688	-5%
Operating expenses	-21,691	-21,706	0%	-24,729	-24,845	0%	-27,229	-27,463	-1%
Operating income	50,110	50,589	-1%	83,894	87,924	-5%	114,057	121,225	-6%
Non-operating income	2,121	2,722	-22%	1,809	3,425	-47%	1,828	3,844	-52%
Pre-tax income	52,231	53,311	-2%	85,703	91,349	-6%	115,885	125,069	-7%
Income tax	-11,428	-11,660	-2%	-19,614	-20,891	-6%	-26,581	-28,687	-7%
Net income	40,652	41,505	-2%	65,937	70,311	-6%	89,152	96,236	-7%
EPS (NT\$)	19.79	20.21	-2%	32.11	34.24	-6%	43.41	46.86	-7%
Margins (%)			ppt			ppt			ppt
Gross margin	39.3%	39.6%	-0.3	43.2%	39.5%	3.6	46.1%	48.5%	-2.4
Operating margin	27.4%	27.7%	-0.3	33.4%	27.0%	6.4	37.2%	39.5%	-2.3
Pre-tax margin	28.6%	29.2%	-0.6	34.1%	29.0%	5.1	37.8%	40.8%	-3.0
Net margin	22.2%	22.7%	-0.5	26.2%	22.2%	4.1	29.1%	31.4%	-2.3

e=Morgan Stanley Research estimates; Source: Company data, Morgan Stanley Research

Price Target Discussion and Valuation Methodology

We lower our 12-month price target to NT\$1,050 (from NT\$1,465): This reflects our lower earnings estimates for CY26-28. Our price target is our base case value, derived from our residual income valuation model. Similar to our analysis of other tech hardware companies within our coverage, we think this methodology provides the most accurate value of the firm by taking into account the cost of equity.

Our key assumptions are:

- Cost of equity of 7.8%:
 - Risk-free rate of 1.0% (10-year Taiwanese government bond yield);
 - Equity risk premium of 6.8%;
 - Beta of 1.1;

- Medium-term growth rate of 12% (down from 16%);
- Terminal growth rate of 3%.

Our price target implies 33x 2027e P/E and 24x 2028e P/E. These are above its five-year historical range of 8x to 18x, which we view as justified considering its longer-term transformation story to become a global total solutions provider, moving from passive into active components. In addition, compared to global passive component peers, we believe Yageo stock remains undervalued with higher ROE and margins but lower P/E vs. Japanese peers (which trade on an average at 40x+ P/E).

Our bull and bear case scenario values decrease by magnitudes similar to the change to our base case value, to NT\$1,490 and NT\$480, respectively.

Exhibit 30: Yageo: Residual Income (RI) Model

	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E
Total Equity	217,162	257,880	306,624	352,983	404,905	463,058	528,189	601,136	682,837	774,341	859,089
Net Profit	40,652	65,937	89,152	99,850	111,832	125,252	140,283	157,117	175,971	197,087	203,000
Return on Equity	20.8%	27.8%	31.6%	30.3%	29.5%	28.9%	28.3%	27.8%	27.4%	27.1%	24.9%
Beta	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Equity Risk Premium (Rm-Rf)	6.8%	6.8%	6.8%	6.8%	6.8%	6.8%	6.8%	6.8%	6.8%	6.8%	6.8%
Risk Free Rate (Rf)	1.0%	1.0%	1.0%	1.0%	1.0%	1.0%	1.0%	1.0%	1.0%	1.0%	1.0%
Cost of Equity	7.8%	7.8%	7.8%	7.8%	7.8%	7.8%	7.8%	7.8%	7.8%	7.8%	7.8%
Residual Income	22,664	43,434	61,443	69,039	76,779	85,440	95,132	105,980	118,125	131,723	132,378
Spread	13.0%	20.0%	23.8%	22.5%	21.8%	21.1%	20.5%	20.1%	19.7%	19.3%	17.1%
Beginning Equity Capital	217,162										
PV of Forecast Period	582,581										
PV of Continuing Value	1,356,663										
Equity Value	2,156,406										
No. of Shares	2,054										
Projected Price	1,050.0										
Implied 2027e PE	33x										
Implied 2028e PE	24x										

E=Morgan Stanley Research estimates; Source: Morgan Stanley Research

Exhibit 31: Yageo: Quarterly earnings estimates

	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26E	4Q26E	2025	2026E	2027E	2028E
Net sales	31,104	32,771	33,087	35,968	38,166	44,456	48,552	51,695	132,930	182,769	251,501	306,659
COGS	(20,018)	(21,119)	(21,112)	(22,551)	(23,624)	(27,340)	(29,305)	(30,699)	(84,800)	(110,968)	(142,878)	(165,373)
Gross profit	11,086	11,652	11,974	13,417	14,542	17,116	19,247	20,896	48,130	71,801	108,623	141,286
Operating expenses	(4,623)	(4,809)	(4,412)	(4,685)	(4,929)	(5,368)	(5,633)	(5,761)	(18,329)	(21,691)	(24,729)	(27,229)
SG&A	(3,758)	(3,774)	(3,578)	(3,800)	(4,035)	(4,461)	(4,681)	(4,800)	(14,910)	(17,977)	(20,687)	(22,876)
R&D	(865)	(834)	(834)	(886)	(894)	(907)	(952)	(961)	(3,419)	(3,714)	(4,042)	(4,354)
Operating income	6,463	7,044	7,562	8,732	9,613	11,749	13,614	15,135	29,801	50,110	83,894	114,057
Non-operating income	710	(15)	650	(26)	712	474	468	468	1,319	2,121	1,809	1,828
Interest income	533	443	379	216	303	345	345	345	1,571	1,339	1,382	1,382
Investment income	117	24	144	195	181	202	222	222	480	827	827	846
Disposal of investment	100	0	0	0	7	0	0	0	100	7	0	0
Exchange gain	(148)	194	(110)	(677)	(139)	(194)	0	0	(742)	(333)	0	0
Other	108	(676)	237	241	359	121	(100)	(100)	(90)	280	(400)	(400)
Pre-tax income	7,173	7,028	8,212	8,706	10,325	12,222	14,081	15,602	31,120	52,231	85,703	115,885
Income tax	(1,610)	(2,016)	(1,817)	(1,900)	(2,287)	(2,766)	(3,098)	(3,276)	(7,343)	(11,428)	(19,614)	(26,581)
Minority interest	33	14	39	56	37	38	38	38	143	151	152	152
Net income	5,530	4,998	6,356	6,751	8,001	9,418	10,945	12,288	23,634	40,652	65,937	89,152
Adj. wtd. avg. shrs (mn)	2,053	2,053	2,054	2,053	2,054	2,054	2,054	2,054	2,053	2,054	2,054	2,054
EPS	2.69	2.43	3.10	3.29	3.90	4.59	5.33	5.98	11.51	19.79	32.11	43.41
Fully Diluted shares (mn)	2,459	2,456	2,055	2,053	2,054	2,054	2,054	2,054	2,053	2,054	2,054	2,054
Diluted EPS (NT\$)	2.25	2.03	3.09	3.29	3.90	4.59	5.33	5.98	11.51	19.79	32.11	43.41
Margins (%)												
Gross Margin	35.6	35.6	36.2	37.3	38.1	38.5	39.6	40.5	36.2	39.3	43.2	46.1
Operating Margin	20.8	21.5	22.9	24.3	25.2	26.4	28.0	29.3	22.4	27.4	33.4	37.2
Pretax Margin	23.1	21.4	24.8	24.2	27.1	27.5	29.0	30.2	23.4	28.6	34.1	37.8
Net Margin	17.8	15.3	19.2	18.8	21.0	21.2	22.5	23.8	17.8	22.2	26.2	29.1
QoQ Growth (%)												
Sales	4	5	1	9	6	16	9	6				
Gross Profit	11	5	3	12	8	18	12	9				
Operating Profit	20	9	7	15	10	22	16	11				
Pretax Profit	15	-2	17	6	19	18	15	11				
Net Profit	49	-10	27	6	19	18	16	12				
YoY Growth (%)												
Sales	9	4	4	20	23	36	47	43	9	37	38	22
Gross Profit	15	6	7	35	31	47	61	56	15	49	51	30
Operating Profit	31	9	14	63	49	67	80	73	27	68	67	36
Pretax Profit	18	-5	15	39	44	74	71	79	16	68	64	35
Net Profit	21	-8	13	82	45	88	72	82	22	72	62	35

Source: Company data, Morgan Stanley Research; E=Morgan Stanley Research estimates.

Exhibit 32: Yageo: Consolidated Financial Summary

Consolidated Income Statement					Consolidated Cash Flow Statement				
NT\$mn (Year End Dec 31)	2025	2026E	2027E	2028E	NT\$mn (Year End Dec 31)	2025	2026E	2027E	2028E
Net sales	132,930	182,769	251,501	306,659	Operating Cashflow	30,863	45,519	65,397	87,035
COGS	(84,800)	(110,968)	(142,878)	(165,373)	Net Profits	23,634	40,652	65,937	89,152
Gross profit	48,130	71,801	108,623	141,286	Depreciation & Amort.	9,834	9,469	9,163	8,866
Operating expenses	(18,329)	(21,691)	(24,729)	(27,229)	Investment losses/(income)	(581)	(827)	(827)	(846)
Operating income	29,801	50,110	83,894	114,057	Working capital change	(12,663)	(840)	(4,837)	(6,896)
Non-operating income	1,319	2,121	1,809	1,828	Other adjustments	10,944	(2,929)	(4,039)	(3,241)
Interest income	1,571	1,339	1,382	1,382	Investing Cashflow	(3,725)	(7,356)	(7,114)	(6,920)
Investment income	480	827	827	846	Capex	(6,056)	(8,000)	(8,000)	(8,000)
Disposal of investment	100	7	-	-	Change of L/T investment	(4,146)	0	0	0
Exchange gain	(742)	(333)	-	-	Change of S/T investment	817	0	0	0
Other	(90)	280	(400)	(400)	Other adjustments	5,191	0	0	0
Pre-tax income	31,120	52,231	85,703	115,885	Financing Cashflow	(3,135)	10,685	(13,435)	(30,967)
Income tax	(7,343)	(11,428)	(19,614)	(26,581)	Increase in L/T debt	(543)	(1,000)	(1,000)	(1,000)
Minority interests	143	151	152	152	Increase in S/T debt	7,613	0	0	0
Net income	23,634	40,652	65,937	89,152	Issuance of stock	0	15,337	0	0
Adj. wtd. Avg. shrs (m)	2,053	2,054	2,054	2,054	Cash dividends	(10,265)	(12,322)	(24,392)	(39,562)
Reported EPS (NT\$)	11.51	19.79	32.11	43.41	Other adjustments	61	8,670	11,957	9,595
Diluted shrs (m)	2,053	2,054	2,054	2,054	FX adjustment	(3,647)	333	0	0
Diluted EPS (NT\$)	11.51	19.79	32.11	43.41	Net change in cash	20,356	49,180	44,848	49,147

Consolidated Balance Sheet					Consolidated Financial Ratios				
NT\$mn (Year End Dec 31)	2025	2026E	2027E	2028E		2025	2026E	2027E	2028E
Cash	81,473	130,653	175,501	224,649	Margins (%)				
Mkt securities	16,634	16,634	16,634	16,634	Gross margin	36.2	39.3	43.2	46.1
AR/RR	27,792	34,112	46,526	59,743	Operating margin	22.4	27.4	33.4	37.2
Inventory	31,636	41,548	52,742	59,653	Pretax margin	23.4	28.6	34.1	37.8
Others	7,811	10,739	14,778	18,019	Net margin	17.8	22.2	26.2	29.1
Current Assets	165,346	233,687	306,182	378,698	YoY growth (%)				
Long-term investments	45,572	45,572	45,572	45,572	Sales	9.3	37.5	37.6	21.9
Fixed assets	65,305	63,192	61,143	59,197	Operating profits	27.4	68.2	67.4	36.0
Other assets	114,566	114,566	114,566	114,566	Pretax profits	15.8	67.8	64.1	35.2
Total Assets	390,789	457,018	527,463	598,034	Net profits	22.1	72.0	62.2	35.2
S/T borrowings	86,196	86,196	86,196	86,196	Others				
AP/PP	18,031	23,596	30,381	35,164	Cash dividend payout (%)	52%	60%	60%	60%
Other ST liabilities	31,852	41,680	53,666	62,115	Cash div (NT\$)	6.00	11.88	19.26	26.05
Total Current Liabilities	136,079	151,472	170,243	183,476	Yield (%)	1%	2%	3%	5%
L/T debt	57,589	56,589	55,589	54,589	Net Debt/Equity (%)	26%	-2%	-20%	-33%
Other LT liabilities	23,124	31,794	43,751	53,346	Liabilities/Equity (%)	125%	110%	105%	95%
Total Liabilities	216,793	239,856	269,583	291,411	Liabilities/Assets (%)	55%	52%	51%	49%
Common shares	5,182	20,519	20,519	20,519	ROAE (%)	14%	21%	28%	32%
Retained Earnings	104,669	132,498	173,216	221,959	ROAA (%)	6%	10%	13%	16%
Other SH' Equity	64,145	64,145	64,145	64,145	AR/RR Turnover (days)	68	62	59	63
Total Shareholders' Equity	173,997	217,162	257,880	306,624	Inventory Turnover (days)	128	120	120	124
Total Liab./SH's Equity	390,789	457,018	527,463	598,034	AP/PP Turnover (days)	71	68	69	72
					Cash conversion cycle	125	114	110	115

Source: Company data, Morgan Stanley Research; E=Morgan Stanley Research estimates.

Risk Reward – Yageo Corp. (2327.TW)

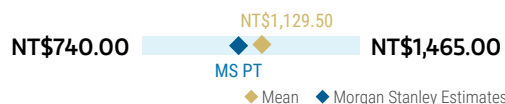
Demand looks stronger and price increases appear likely - stay OW

PRICE TARGET NT\$1,050.00

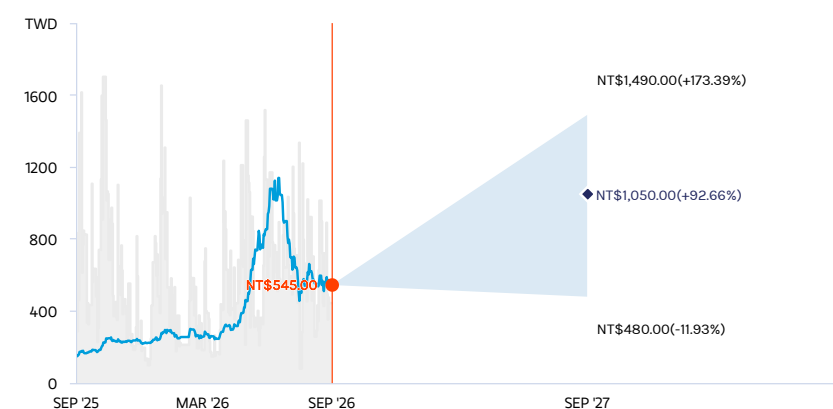
Base case, multistage residual income valuation model. Key assumptions: cost of equity 7.8% (risk-free rate 1%, equity risk premium 6.8% and beta of 1.1), medium-term growth rate 12%, terminal growth rate 3%.

Consensus Price Target Distribution

Source: Refinitiv, Morgan Stanley Research



RISK REWARD CHART



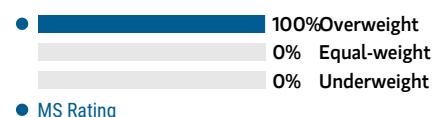
Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- There is fear that leading Japanese and Korean MLCC suppliers are shifting more capacity to cater to the high end, resulting in undersupply for the midrange/low end.
- The same narrative that drove the 2018 cycle is now back in play.
- Utilization rates are increasing, so suppliers are expecting price hikes for the midrange/low end. Distributors and ODMs are both building inventory.
- AI demand is growing and tantalum capacitor pricing may show further price increases in 2H.
- All of this should help support a re-rating in the stock.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Electric Vehicles:	Positive
Self-help:	Positive
Special Situation:	Positive

View descriptions of Risk Rewards Themes [here](#)

BULL CASE

NT\$1,490.00

34x 2028e EPS

Demand continues to grow faster than expected in 2027-28: 1) Demand improves significantly, driving pricing power for Yageo in premium and standard products alike. 2) Yageo makes meaningful progress into AI, driving significant share gains. 3) Yageo takes on more acquisition initiatives to further widen its product portfolio. 4) Severe undersupply leads to continued price hikes for an extended period of time, which helps expand Yageo's margins in 2027-28.

BASE CASE

NT\$1,050.00

24x 2028e EPS

Demand remains strong into 2027-28: 1) Utilization remains high for premium products. 2) Utilization for standard products stabilizes at 70%+ with resilient consumer tech demand. 3) Transformation and synergies are ongoing with continued active initiatives. 4) Yageo continues to drive expansion into the AI realm and slowly gains market share. 5) High utilization rates and inflationary raw material costs result in continued price hikes.

BEAR CASE

NT\$480.00

11x 2028e EPS

Earnings erode as macro conditions weaken and demand falls again in 2027-28: 1) Non-AI demand weakens from 2H26 onward, despite AI demand remaining strong. 2) Pricing power shifts to end customers with heightened competition among peers. 3) Acquisitions do not bear fruit, resulting in layoffs and leading to cost burdens. 4) Automotive and industrial end demand does not recover, hampering Yageo's premium business. 5) Yageo does not make any material advancements into AI.

Risk Reward – Yageo Corp. (2327.TW)

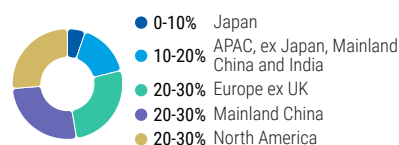
KEY EARNINGS INPUTS

Drivers	Dec 2025	Dec 2026e	Dec 2027e	Dec 2028e
MLCC ASP y/y (%)	0.6	24.3	65.3	14.9
MLCC GM (%)	31.7	36.4	43.6	47.2
r-chip ASP y/y (%)	0.8	11.8	47.0	27.3
r-chip GM (%)	43.4	45.5	50.3	53.4

INVESTMENT DRIVERS

- Transformation into global total solutions service provider
- Premium/standard sales mix
- Utilization rates and capacity
- Inventory and pricing

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

2/5
MOST 3 Month
Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Faster-than-expected revenue/profit recognition from stronger AI server/notebook demand
- Further price hikes led by stronger pricing power arising from better demand and/or tighter supply

RISKS TO DOWNSIDE

- A sudden bursting of demand for AI
- Inventory risk
- Consumer electronics demand risk

OWNERSHIP POSITIONING

Inst. Owners, % Active 70.4%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2027e

Sales / Revenue (NT\$, mn) 212,000 251,501 263,812
243,261

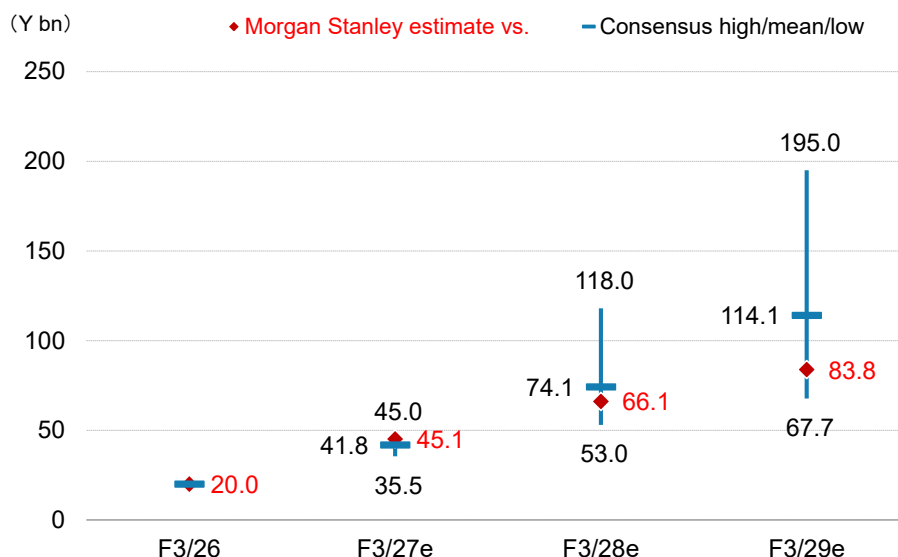
EBITDA (NT\$, mn) 72,235 93,057 107,728
91,475

Net income (NT\$, mn) 50,264 65,937 73,325
64,949

EPS (NT\$) 24.48 32.11 35.40
31.45

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Exhibit 35: Taiyo Yuden: OP forecasts comparison

Note: Consensus is based on FactSet; Source: Fact Set, Morgan Stanley Research

Taiyo Yuden share price risk/reward: We upgrade Taiyo Yuden from UW to EW, as we believe its share price, having fallen 56.0% from its July 1 peak, has limited downside risk and offers 6% upside over the next 12-18 months. We lower our DCF-based price target from ¥12,500 to ¥10,500. This mainly reflects an increase in the DCF model's risk-free rate from 2.60% to 3.0%. We rate the stock EW rather than OW because: (1) In its core MLCC business, we estimate Taiyo Yuden's MLCC market share was 11.3% in 2025, down from 12.7% in 2020, and we think this downtrend continued into 2026, with its market share shrinking to 10.7% in 1Q and 9.9% in 2Q. (2) We believe Taiyo Yuden is lagging behind Murata (No. 1 share) and SEMCO (No. 2) in small, high-capacity MLCCs for AI server accelerator boards, where demand is rapidly expanding. (3) We expect consensus estimates for F3/28 onward to be revised down after 2Q results.

Bull case ¥14,000, 49% projected upside

In our bull case, F3/27 sales of MLCCs for AI servers/data centers are 7% higher than in our base case, for OP of ¥56.5bn in F3/27, ¥85.5bn in F3/28, and ¥176.3bn in F3/31. Our forex assumptions and DCF assumptions match our base case. Bull case fair value corresponds to 50.7x our fully-diluted F3/27 EPS estimate of ¥276.1, 33.6x our fully-diluted F3/28 EPS estimate of ¥417.0, and 15.9x our full-diluted F3/31 EPS estimate of ¥879.1.

Bear case ¥4,500

Taiyo Yuden's core products (MLCCs and inductors) are used in all kinds of electronic devices, and are easily affected by the macro economy and inventory cycle. In our bear case, we assume slower global economic growth rates and lower production of high-end consumer goods like high-end smartphones than in our base case, for OP of ¥39.2bn in F3/27, ¥56.1bn in F3/28, and ¥65.8bn in F3/31. Our forex assumptions and DCF assumptions are the same as in our base case. Here we look for ROE of 7.3% in F3/27 and 10.1% in F3/28, improving to a level above capital cost of 7.9%. With profit below market expectations, we assume the share price falls to a P/B of 1.8x.

Risk Reward – Taiyo Yuden (6976.T)

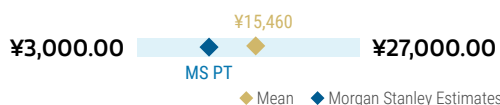
Share price has fallen, but upside potential likely to remain limited

PRICE TARGET ¥10,500

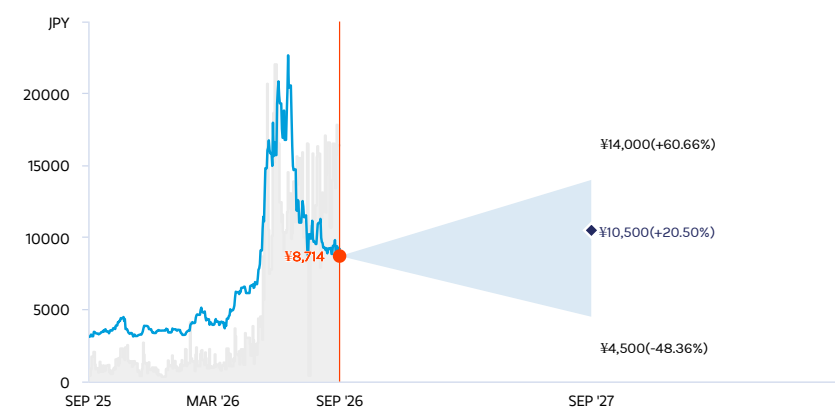
Derived from the base case. DCF assumptions: 3.0% risk-free rate, 1.54 equity beta, 3.2% risk premium, for 7.4% WACC; zero growth from F3/36.

Consensus Price Target Distribution

Source: Refinitiv, Morgan Stanley Research



RISK REWARD CHART



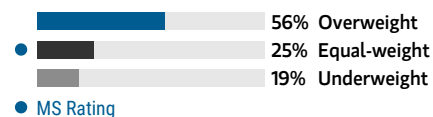
Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

EQUAL-WEIGHT THESIS

- While we expect earnings growth to continue, we believe the upside potential for the stock is likely to be limited.
- We expect continued earnings growth in the core MLCC business..
- We expect the contribution from premium MLCC products for AI/data center applications to remain limited.
- We expect sales growth for power inductors used in 5G smartphones and base stations to continue.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

New Data Era: *Positive*

View descriptions of Risk Rewards Themes [here](#)

BULL CASE

¥14,000

Derived from DCF

MLCC sales exceed the base case. OP ¥56.5bn in F3/27, ¥85.5bn in F3/28, ¥176.3bn in F3/31. Same FX & DCF model assumptions as the base case. Fair value equates to P/E 50.7x on F3/27e diluted EPS of ¥276.1, 33.6x on F3/28e diluted EPS of ¥417.0, and 15.9x on F3/31e diluted EPS of ¥879.1.

BASE CASE

¥10,500

Derived from DCF

Earnings are expanding driven by growth in MLCC sales, but market expectations appear overly high: OP ¥45.1bn in F3/27, ¥66.1bn in F3/28, ¥107.7bn in F3/31. FX assumption: ¥155/US\$. Base case fair value equates to P/E 48.1x on F3/27e diluted EPS of ¥218.3, 33.0x on F3/28e diluted EPS of ¥318.5, and 19.8x on F3/31e diluted EPS of ¥530.1.

BEAR CASE

¥4,500

F3/27e diluted BPS ¥2,552 x P/B 1.8

Stays competitive in core products, but slowing end demand drags profits down. OP ¥39.2bn in F3/27, OP ¥56.1bn in F3/28, ¥65.8bn in F3/31. Same FX and DCF assumptions as in the base case. ROE 7.3% in F3/27, 10.1% in F3/28, improving from 4.5% in F3/26 and higher in F3/28 than cost of capital (7.9%), but with earnings lower than market expectation, P/B falls to 1.8x.

Risk Reward – Taiyo Yuden (6976.T)

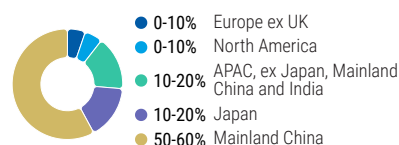
KEY EARNINGS INPUTS

Drivers	Mar 2026	Mar 2027e	Mar 2028e	Mar 2029e
Total Operating Profit (¥, bn)	20	45	66	84
Capacitors Sales YoY (%)	8.5	27.4	16.6	18.0
Complex Devices Sales YoY (%)	(35.6)	(18.3)	19.9	0.0
SG&A Expenses (¥, bn)	61.9	67.9	69.6	71.8
Capex (¥, bn)	40.4	42.0	60.0	60.0

INVESTMENT DRIVERS

- Continuing tech advances via smaller/higher-capacitance MLCCs, and Taiyo Yuden's strong tech competitiveness.
- We expect sales of premium MLCCs for AI/data center applications to remain limited.

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

5/5
MOST

3 Month
Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Increase in premium MLCC sales for AI/DC use.
- Power inductor sales for 5G handsets and others grow steeper than expected.

RISKS TO DOWNSIDE

- We estimate a ¥1/US\$ depreciation affects annual OP increases by ¥0.9bn.
- Since MLCCs are fitted in various electronic devices, demand depends to some extent on the rate of global economic growth, and prices can swing widely depending on supply/demand balances.

OWNERSHIP POSITIONING

Inst. Owners, % Active **80.5%**

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Mar 2027e

Sales / Revenue (¥, bn) 384.0 **407.7** **428.7**

EBITDA (¥, bn) 86.0 **91.8** **95**

Net income (¥, bn) 24.0 **28.6** **30**

EPS (¥) 175.8 **218.3** **223.8**

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Risk Reward Reference links

1. View explanation of Options Probabilities methodology - [Options_Probabilities_Exhibit_Link.pdf](#)
2. View descriptions of Risk Rewards Themes - [RR_Themes_Exhibit_Link.pdf](#)
3. View explanation of regional hierarchies - [GEG_Exhibit_Link.pdf](#)
4. View explanation of Theme/Exposure methodology - [ESG_Sustainable_Solutions_External_Link.pdf](#)
5. View explanation of HERS methodology - [ESG_HERS_External_Link.pdf](#)

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(as of August 31, 2026)

The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.

For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1555	42%	450	48%	29%	767	44%
Equal-weight/Hold	1577	43%	395	42%	25%	770	44%
Not-Rated/Hold	3	0%	1	0%	33%	1	0%
Underweight/Sell	559	15%	93	10%	17%	216	12%
Total	3,694		939			1754	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

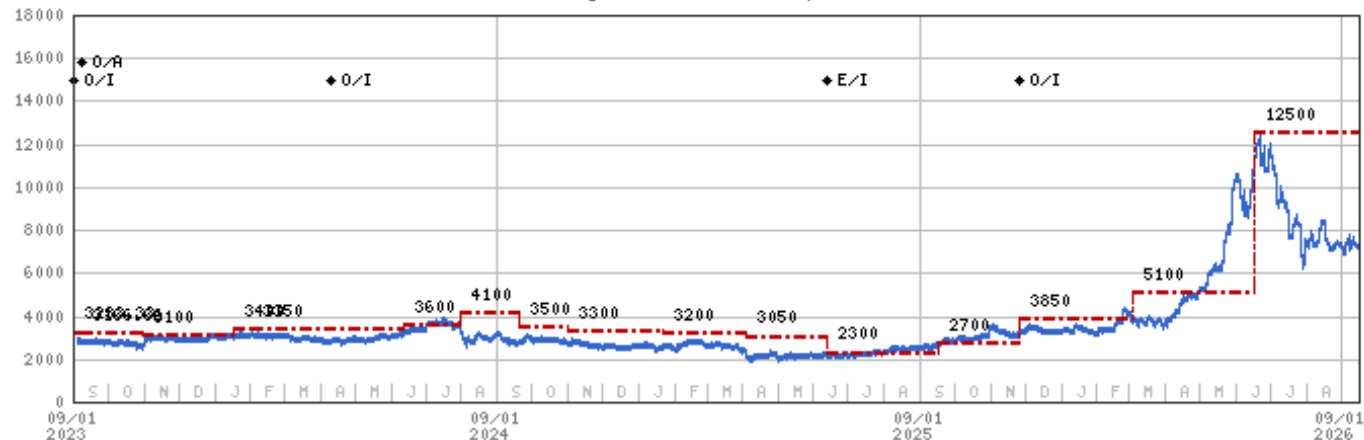
In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Stock Price, Price Target and Rating History (See Rating Definitions)

Murata Manufacturing (6981.T) - As of 09/15/26 GMT in JPY
Industry : Electronic Components



Stock Rating History: 9/1/21 : O/I; 10/19/21 : E/I; 4/11/22 : O/I; 9/8/23 : O/A; 4/10/24 : O/I; 6/13/25 : E/I; 11/26/25 : O/I

Price Target History: 7/30/21 : 3833.33; 10/19/21 : 3166.66; 1/17/22 : 3233.33; 4/11/22 : 3333.33; 6/30/22 : 3100; 7/28/22 : 3066.66; 11/1/22 : 2833.33; 1/17/23 : 2733.33; 2/2/23 : 2700; 3/22/23 : 2900; 6/7/23 : 3166.66; 7/31/23 : 3233.33; 9/8/23 : 3166.66; 10/31/23 : 3100; 1/17/24 : 3400; 2/2/24 : 3350; 6/12/24 : 3600; 7/31/24 : 4100; 9/20/24 : 3500; 11/1/24 : 3300; 1/22/25 : 3200; 4/4/25 : 3050; 6/13/25 : 2300; 9/17/25 : 2700; 11/26/25 : 3850; 3/4/26 : 5100; 6/17/26 : 12500

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target — No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —

Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

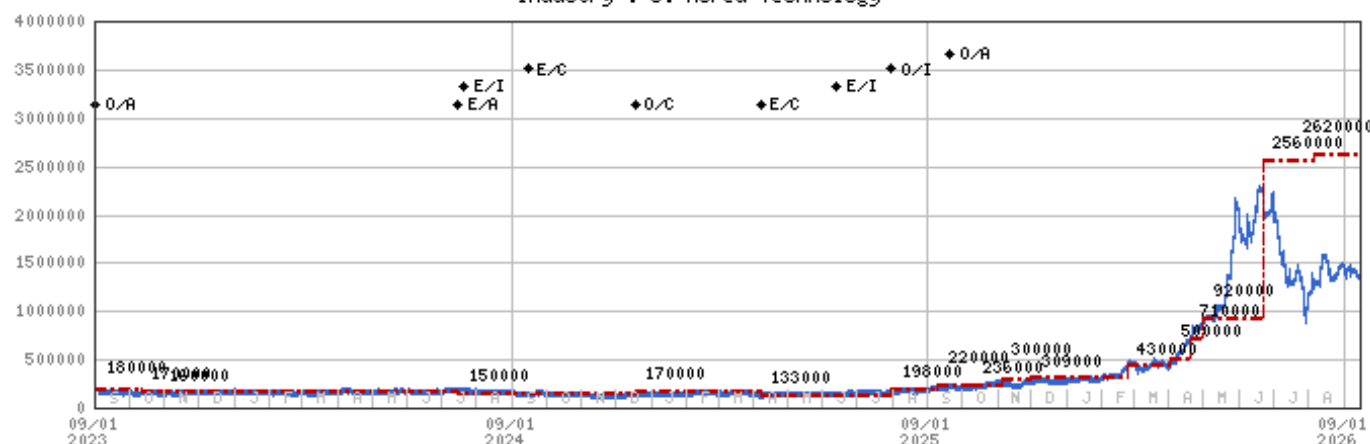
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Samsung Electro-Mechanics (009150.KS) - As of 09/15/26 GMT in KRW
Industry : S. Korea Technology



Stock Rating History: 9/1/21 : E/C; 6/1/22 : U/C; 10/4/22 : U/A; 11/9/22 : O/A; 7/15/24 : E/A; 7/21/24 : E/I; 9/15/24 : E/C; 12/18/24 : O/C; 4/7/25 : E/C; 6/13/25 : E/I; 7/31/25 : O/I; 9/21/25 : O/A

Price Target History: 5/12/21 : 180000; 10/19/21 : 170000; 10/27/21 : 160000; 1/24/22 : 180000; 3/18/22 : 170000; 6/1/22 : 140000; 11/9/22 : 180000; 10/11/23 : 170000; 10/27/23 : 160000; 7/15/24 : 150000; 12/18/24 : 170000; 4/7/25 : 133000; 7/31/25 : 198000; 9/10/25 : 220000; 10/10/25 : 236000; 11/5/25 : 300000; 12/1/25 : 309000; 2/23/26 : 430000; 4/2/26 : 500000; 4/19/26 : 710000; 5/1/26 : 920000; 6/22/26 : 2560000; 8/7/26 : 2620000

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target — No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —

Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

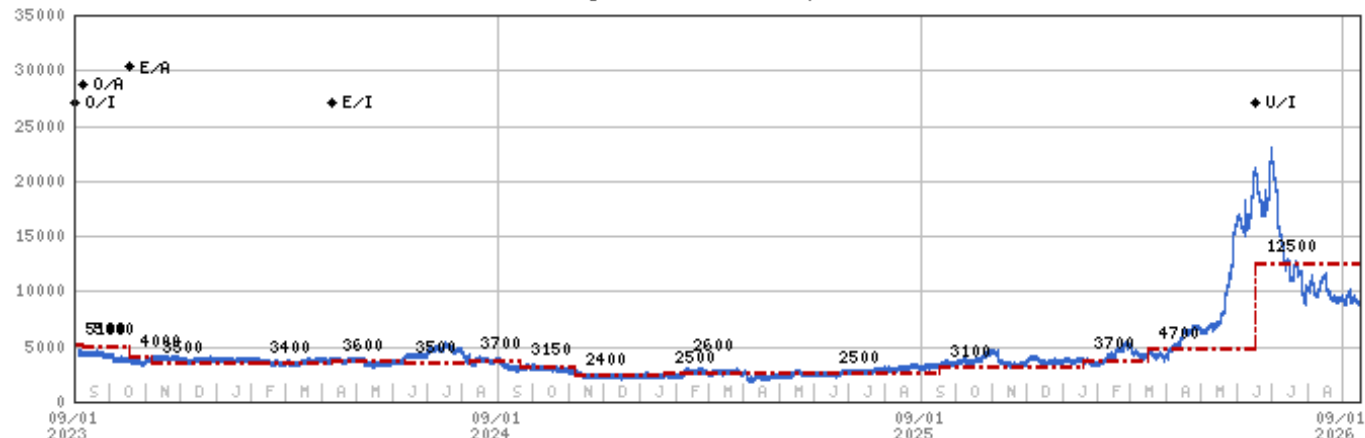
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Taiyo Yuden (6976.T) - As of 09/15/26 GMT in JPY
Industry : Electronic Components



Stock Rating History: 9/1/21 : E/I; 10/19/21 : U/I; 4/11/22 : E/I; 1/17/23 : O/I; 9/8/23 : O/A; 10/19/23 : E/A; 4/10/24 : E/I; 6/17/26 : U/I

Price Target History: 8/5/21 : 6400; 10/19/21 : 4900; 1/17/22 : 5200; 2/5/22 : 5000; 4/11/22 : 6200; 6/30/22 : 5300; 8/3/22 : 4900; 9/28/22 : 4500; 11/7/22 : 4100; 1/17/23 : 5000; 2/7/23 : 4900; 3/22/23 : 5000; 6/7/23 : 5300; 8/3/23 : 5100; 9/8/23 : 5000; 10/19/23 : 4000; 11/7/23 : 3500; 2/7/24 : 3400; 4/10/24 : 3600; 6/12/24 : 3500; 8/7/24 : 3700; 9/20/24 : 3150; 11/7/24 : 2400; 1/22/25 : 2500; 2/7/25 : 2600; 6/13/25 : 2500; 9/17/25 : 3100; 1/19/26 : 3700; 3/17/26 : 4700; 6/17/26 : 12500

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target — No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —

Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

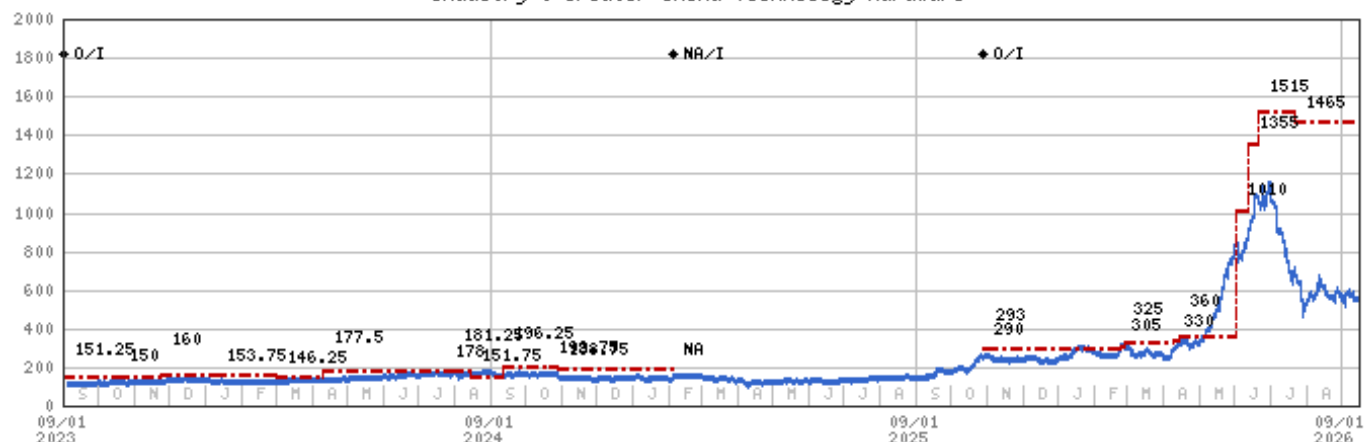
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Yageo Corp. (2327.TW) - As of 09/15/26 GMT in TWD
Industry : Greater China Technology Hardware



Stock Rating History: 9/1/21 : E/I; 1/4/22 : O/I; 2/5/25 : NA/I; 10/28/25 : O/I

Price Target History: 7/27/21 : 182.18; 10/19/21 : 141.35; 1/4/22 : 188.46; 6/23/22 : 155.48; 8/12/22 : 144.49; 10/28/22 : 116.25; 11/10/22 : 123.75; 1/9/23 : 157.5; 4/12/23 : 158.75; 7/17/23 : 152.5; 7/26/23 : 151.25; 10/18/23 : 150; 11/23/23 : 160; 1/9/24 : 153.75; 3/1/24 : 146.25; 4/11/24 : 177.5; 7/23/24 : 178; 7/30/24 : 181.25; 8/15/24 : 151.75; 9/12/24 : 196.25; 10/21/24 : 193.75; 10/29/24 : 188.75; 2/5/25 : NA; 10/28/25 : 290; 10/31/25 : 293; 2/23/26 : 305; 2/26/26 : 325; 4/9/26 : 330; 4/15/26 : 360; 6/2/26 : 1010; 6/13/26 : 1355; 6/22/26 : 1515; 7/22/26 : 1465

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target -- No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —
Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

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